

• Patient name: **Laura Senne Moretti**

• Gender: **Female**

• Date of birth: **04-08-1994**

• Sample code: **BOX03464AANUT**

• Sample date: **09-06-2026**

• Date of the results: **15-06-2026**



Fagron NutriGen

Doctor report



Fagron NutriGen

CONTENT

1. Patient identification data

2. Patient characteristics

3. Genetic results overview

- 3.1. Morphological genetics for weight control
- 3.2. Behavioural genetics in food intake
- 3.3. Flavour sensitivities
- 3.4. Fat metabolism
- 3.5. Lipid metabolism
- 3.6. Carbohydrate metabolism
- 3.7. Glucose metabolism
- 3.8. Efficacy of exercise
- 3.9. Detoxification imbalances
- 3.10. Intolerance
- 3.11. Hormones
- 3.12. Inflammation
- 3.13. Vitamins
- 3.14. Minerals
- 3.15. Effectiveness of diets

4. Recommended nutritional plan

5. Recommended supplements

6. Recommended formulations



Fagron NutriGen

CONTENT

7. Complete genetic results

- 7.1. Morphological genetics for weight control
 - 7.1.1. Genetic risk of overweight/obesity
 - 7.1.2. Risk of rebound weight gain
 - 7.1.3. Risk of increased BMI
 - 7.1.4. Basal metabolic rate (burn calories at rest)
 - 7.1.5. Weight loss capability during diet interventions
- 7.2. Behavioural genetics in food intake
 - 7.2.1. Appetite and anxiety risk
 - 7.2.2. Satiety: Feeling Full
- 7.3. Flavour sensitivities
 - 7.3.1. Bitter taste sensitivity
 - 7.3.2. Salt sensitivity
 - 7.3.3. Sweet flavour preference
- 7.4. Fat metabolism
 - 7.4.1. Response to monounsaturated fats (MUFAs)
 - 7.4.2. Response to polyunsaturated fats (PUFAs)
 - 7.4.3. Response to fat intake to improve the HDL levels
- 7.5. Lipid metabolism
 - 7.5.1. Predisposition to reduced HDL levels
 - 7.5.2. Predisposition to increased levels of triglycerides
 - 7.5.3. Predisposition to increased oxidation of LDL
 - 7.5.4. Risk of increased cholesterol LDL levels
 - 7.5.5. Risk of unbalanced Triglycerides/HDL ratio
- 7.6. Carbohydrate metabolism
 - 7.6.1. Capability to digest starchy food
 - 7.6.2. Refined carbohydrate sensitivity
 - 7.6.3. Carbohydrates and HDL levels predisposition
 - 7.6.4. Carbohydrates and LDL levels
- 7.7. Glucose metabolism
 - 7.7.1. Risk of increased glucose levels in plasma after fasting
 - 7.7.2. Risk of insulin resistance
 - 7.7.3. Risk of Type-II diabetes
- 7.8. Efficacy of exercise
 - 7.8.1. Benefits from endurance exercise for improving HDL levels
 - 7.8.2. Exercise to reduce body fat
- 7.9. Detoxification imbalances
 - 7.9.1. Antioxidant capability
- 7.10. Intolerance
 - 7.10.1. Lactose intolerance risk
 - 7.10.2. Risk of celiac disease
 - 7.10.3. Fructose intolerance risk
 - 7.10.4. Caffeine metabolism
 - 7.10.5. Alcohol metabolism
- 7.11. Hormones
 - 7.11.1. Leptin
 - 7.11.2. Visfatin
 - 7.11.3. Ghrelin
 - 7.11.4. Adiponectin
- 7.12. Inflammation
 - 7.12.1. TNF- α
 - 7.12.2. IL-6
 - 7.12.3. IL-10
- 7.13. Vitamins
 - 7.13.1. Vitamin A
 - 7.13.2. Vitamin B6
 - 7.13.3. Vitamin B9 (folate)
 - 7.13.4. Vitamin B12
 - 7.13.5. Vitamin C
 - 7.13.6. Vitamin D
 - 7.13.7. Vitamin E
- 7.14. Minerals
 - 7.14.1. Calcium malabsorption risk
 - 7.14.2. Predisposition to dysregulated calcium levels
 - 7.14.3. Risk of iron overload
 - 7.14.4. Risk of low iron plasma levels
 - 7.14.5. Predisposition to dysregulated magnesium levels
 - 7.14.6. Predisposition to dysregulated selenium levels
 - 7.14.7. Sodium sensitivity
- 7.15. Effectiveness of diets
 - 7.15.1. Efficacy of low calorie diets
 - 7.15.2. Efficacy of low carbohydrate diets
 - 7.15.3. Efficacy of low fat diets

8. Methodology

9. References

Patient identification data

1

Patient's name and last name —●— **Laura Senne Moretti**
Date of Birth —●— **04-08-1994**
Gender —●— **Female**
Sample type —●— **Buccal mucosa**
Sample code —●— **BOX03464AANUT**
Sample date —●— **09-06-2026**
Report date —●— **16-06-2026**

Doctor's name and last name —●— **Ketone Clinical Laboratory**
Email address —●— **info@ketone.ae**
Telephone —●— **+971 4 892 2499**

LEGAL DISCLAIMER

Fagron Genomics, S.L.U carries out genetic tests upon request by healthcare professionals, in relation to biological samples from patients obtained by the healthcare professional. Our tests do not replace a medical consultation, nor do they make up a diagnostic or treatment, nor should they be interpreted this way. Only healthcare professionals can interpret the results of said tests, based on their knowledge of the clinical records of the patients and other relevant factors and, under their responsibility, give a diagnostic or prescribe treatment to the patient. We decline all responsibility derived from the use and interpretation of the results of our tests by the solicitant healthcare professional. Fagron Genomics, S.L.U expressly reserves any legal actions in case of an inappropriate, negligent or incorrect use or interpretation of the results of our tests. It is the responsibility of the healthcare professional who requests a test to guarantee to the patient the appropriate genetic advice as foreseen by Law 14/2007, of 3rd July, of biomedical research. As Fagron Genomics, S.L.U does not have access to the personal identifiable information about the patient from whom the sample comes, it is the responsibility of the requesting healthcare professional to comply with the applicable data protection Laws and regulations.



Fagron Genomics,
SRN: ES-MF-000001092
C/ de les Cosidores, 150
08226 Terrassa, Barcelona (Spain)



IVDD
Self-Declared
98/79/EC



FGMS-Nutri



In Vitro Diagnostic MedicalDevice



8437024682FGMS-NutriVC

2

Patient characteristics

CGATCGATCGATCGATCGATCGTATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTG
GACCGACAGTCATCCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGATCGATCGATCGA
ATCGATCGATCGATCGATCGACGTACTGATCGATCGAATCGATCGATCGATCGACGTACTGATCGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGAT
ATCGATCGATCGATCGATCGATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTCGATCGATCGATCGAGTACTGAT

Patient characteristics

2

Weight related variables

Age (years) 31

Height (cm) 173

Current weight (kg) 69

Goal weight (kg) 64

Current BMI 23,05

Goal BMI 21,38

Weight type Normoweight

Physical exercise and metabolism related factors

Daily sport activity Light

- Basal metabolism -

Current (cal) 1.455

Target (cal) 1.405

- Current DEE -

Current (Kcal) 2.000

Target (Kcal) 1.932

Variation (Kcal) -69

IMPORTANT

In case of underweight, Obesity Type I, II, III, IV and/or existing pathologies, the results of this test must be evaluated and implemented ONLY by a doctor / endocrinologist / specialist / medical practitioner.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

- 3.1. Morphological genetics for weight control
- 3.2. Behavioural genetics in food intake
- 3.3. Flavour sensitivities
- 3.4. Fat metabolism
- 3.5. Lipid metabolism
- 3.6. Carbohydrate metabolism
- 3.7. Glucose metabolism
- 3.8. Efficacy of exercise
- 3.9. Detoxification imbalances
- 3.10. Intolerance
- 3.11. Hormones
- 3.12. Inflammation
- 3.13. Vitamins
- 3.14. Minerals
- 3.15. Effectiveness of diets

CGATCGATCGATCGATCGATCGTATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTG
GACCGACAGTTCATCCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGATCGATCGATCGA
ATCGATCGATCGATCGATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGAT
ATCGATCGATCGATCGATCGATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGAT
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGAT

Genetic results overview

3

3.1.	CATEGORY	DESCRIPTION	RESULTS
	 Morphological genetics for weight control	Medium-high genetic predisposition to being overweight. In case of overweight or obesity, it is caused mainly by inherited genetics. Following the recommendations of this DNA analysis will improve outcomes. MEDIUM-HIGH RISK OF OVERWEIGHT/OBESITY ● HIGH REBOUND EFFECT ● MEDIUM-LOW RISK OF INCREASED BMI ● HIGH BURNER AT REST ● SLOW WEIGHT LOSS ●	38.6% Pg. 26 Pg. 27 Pg. 28 Pg. 29 Pg. 30
	- Genetic risk of overweight/obesity - Risk of rebound weight gain - Risk of increased BMI - Basal metabolic rate (burn calories at rest) - Weight loss capability during diet interventions		
3.2.	CATEGORY	DESCRIPTION	RESULTS
	 Behavioural genetics in food intake	Medium-high dysregulation of food intake behaviour. High predisposition to being overweight. Strategies to improve satiety should be considered. INCREASED APPETITE AND ANXIETY RISK ● SLIGHTLY LOWER SATIETY ●	46.58% Pg. 31 Pg. 32
	- Appetite and anxiety risk - Satiety: Feeling Full		
3.3.	CATEGORY	DESCRIPTION	RESULTS
	 Flavour sensitivities	Slightly increased flavour sensitivity. No important impact on the diet is predicted. NORMAL BITTER TASTE SENSITIVITY ● MEDIUM-LOW SALT SENSITIVITY ● HIGHLY INCREASED SWEET PREFERENCE ●	51% Pg. 33 Pg. 34 Pg. 35
	- Bitter taste sensitivity - Salt sensitivity - Sweet flavour preference		

INDICATIONS

■ Positive effect
 ■ Medium-positive effect
 ■ Medium-negative effect
 ■ Negative effect

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

3

3.4.	CATEGORY	DESCRIPTION	RESULTS
	 Fat metabolism	Highly negative fat burning capacity. It would be recommended to greatly decrease the fat intake.	16.61% 
	Response to monounsaturated fats (MUFAs)	VERY LOW MUFA METABOLISM ●	Pg. 36
	Response to polyunsaturated fats (PUFAs)	LOW PUFA METABOLISM ●	Pg. 37
	Response to fat intake to improve the HDL levels	VERY LOW BENEFITS TO IMPROVE HDL ●	Pg. 38
3.5.	CATEGORY	DESCRIPTION	RESULTS
	 Lipid metabolism	Affected lipid metabolism. Cholesterol and triglyceride levels may show irregular results in blood analyses. Specific LDL or HDL treatments would be recommended. Increased cardiovascular risk.	48.92% 
	Predisposition to reduced HDL levels	NORMAL HDL LEVELS ●	Pg. 39
	Predisposition to increased levels of triglycerides	HIGHLY INCREASED TRIGLYCERIDES ●	Pg. 40
	Predisposition to increased oxidation of LDL	SLIGHTLY INCREASED LDL OXIDATION ●	Pg. 41
	Risk of increased cholesterol LDL levels	INCREASED LDL LEVELS ●	Pg. 42
	Risk of unbalanced Triglycerides/HDL ratio	SLIGHTLY INCREASED TG/HDL RATIO ●	Pg. 43
3.6.	CATEGORY	DESCRIPTION	RESULTS
	 Carbohydrate metabolism	Highly negative carbohydrate metabolism: Carbohydrate intake will lead to dysregulation in cholesterol levels and also to increased calorie and fat intake. Eliminating refined carbohydrates is urgent; move to wholegrain carbohydrates and reduce the quantity.	19.13% 
	Capability to digest starchy food	INCREASED STARCH DIGESTION ●	Pg. 44
	Refined carbohydrate sensitivity	VERY HIGH CARBOHYDRATE SENSITIVITY ●	Pg. 45
	Carbohydrates and HDL levels predisposition	HIGH RISK OF HDL DYSREGULATION ●	Pg. 46
	Carbohydrates and LDL levels	HIGH RISK OF LDL DYSREGULATION ●	Pg. 47

INDICATIONS

■ Positive effect
 ■ Medium-positive effect
 ■ Medium-negative effect
 ■ Negative effect

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

3

3.7.	CATEGORY	DESCRIPTION	RESULTS
	 Glucose metabolism	Medium-high dysregulation of glucose metabolism. Intake of refined sugar and carbohydrates will be dangerous. High risk of developing Type-II diabetes.	30.36% 
	• Risk of increased glucose levels in plasma after fasting	MEDIUM-HIGH RISK OF HIGH GLUCOSE LEVELS ●	Pg. 48
	• Risk of insulin resistance	MEDIUM-HIGH INSULIN RESISTANCE ●	Pg. 49
	• Risk of Type-II diabetes	MEDIUM-HIGH DIABETES TYPE-II RISK ●	Pg. 50
3.8.	CATEGORY	DESCRIPTION	RESULTS
	 Efficacy of exercise	Low-medium efficacy of exercise to reduce body fat and regulate cholesterol levels. Intensive dietary interventions may be the best option.	25.56% 
	• Benefits from endurance exercise for improving HDL levels	VERY LOW BENEFITS FROM EXERCISE FOR IMPROVING HDL ●	Pg. 51
	• Exercise to reduce body fat	MEDIUM-HIGH BENEFIT FROM EXERCISE TO REDUCE FAT ●	Pg. 52
3.9.	CATEGORY	DESCRIPTION	RESULTS
	 Detoxification imbalances	Slightly reduced detoxification capacities. Try to decrease toxin exposure and intake.	72.19% 
	• Antioxidant capability	SLIGHTLY REDUCED ANTIOXIDANT CAPABILITY ●	Pg. 53
3.10.	CATEGORY	DESCRIPTION	
	 Intolerance	Please find below the different analysed categories related to intolerances and sensitivities.	
	• Lactose intolerance risk	LOWER RISK OF LACTOSE INTOLERANCE ●	Pg. 54
	• Risk of celiac disease	MEDIUM-HIGH RISK OF CELIAC DISEASE ●	Pg. 55
	• Fructose intolerance risk	LOWER RISK OF FRUCTOSE INTOLERANCE ●	Pg. 57
	• Caffeine metabolism	SLOW CAFFEINE METABOLIZER ●	Pg. 58
	• Alcohol metabolism	NORMAL ALCOHOL METABOLISM ●	Pg. 59

INDICATIONS

■ Positive effect ■ Medium-positive effect ■ Medium-negative effect ■ Negative effect

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

3

3.11. Hormones

CATEGORY	DESCRIPTION
 Leptin	Leptin is a hormone which main function is sending a signal to the brain for food intake regulation. Leptin is commonly called the "satiety hormone". Low levels of leptin may imply problems of overeating and/or burning the stored fat. LEP-R is the gene coding for the cellular receptor of the leptin hormone. Its capability to bind leptin and start the cellular signalling is key for the satiety regulation function. Lower leptin binding capability may lead to high possibilities of leptin resistance, overeating and lower fat burning.

• LEP

Predisposition to increased levels of leptin leading to leptin receptor saturation that may lead to increased risk of overeating and lower fat burning.

CATEGORY	DESCRIPTION
 Visfatin	Visfatin is an adipokine with an inflammatory and catabolic profile that has been associated with several metabolic risk factors, such as obesity, insulin resistance, and Type-II diabetes.

• NAMPT

High predisposition to increased levels of circulating visfatin augmenting considerably the risk of an altered inflammatory response.

CATEGORY	DESCRIPTION
 Ghrelin	Ghrelin is a hormone produced in the gut, often termed "the hunger hormone", since it causes an increase in appetite through its effect in the brain. Imbalances in ghrelin are associated with appetite increase, increased calorie consumption and fat storage.

• GHSR

Predisposition to normal ghrelin receptor (GHSR) expression. Circulating levels of this hormone and the sensation of hunger are increased.

CATEGORY	DESCRIPTION
 Adiponectin	Adiponectin is a hormone that regulates glucose levels and fatty acid breakdown. Low levels of adiponectin are associated with inflammation, lipid abnormalities and insulin resistance.

• ADIPOQ

High predisposition to lower adiponectin plasma levels that leads to an increased inflammation process, lipid abnormalities and insulin resistance.

• ADIPOQ

High predisposition to lower adiponectin plasma levels that leads to an increased inflammation process, lipid abnormalities and insulin resistance.

INDICATIONS

 **Positive effect** **Medium-positive effect** **Medium-negative effect** **Negative effect**

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

3

3.12. Inflammation

CATEGORY	DESCRIPTION
 TNF-α	TNF-α is a pro-inflammatory cytokine, strongly linked to many inflammatory conditions, expressed in, and secreted by adipose tissues. Increased levels are associated with obesity-induced inflammation, adiposity and insulin resistance.

• TNF-α

Predisposition to average levels of TNF-alpha.



CATEGORY	DESCRIPTION
 IL-6	IL-6 is an interleukin with mainly pro-inflammatory functions and is commonly used as inflammatory marker. High levels of IL-6 are associated with obesity, insulin resistance and metabolic syndrome.

• IL-6

Predisposition to highly increased levels of IL-6. Pro-inflammation.



CATEGORY	DESCRIPTION
 IL-10	IL-10 is a cytokine with potent anti-inflammatory properties.

• IL-10

Predisposition to higher levels of the anti-inflammatory cytokine IL-10.



INDICATIONS

 **Positive effect**  **Medium-positive effect**  **Medium-negative effect**  **Negative effect**

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

3

3.13. Vitamins

Major genetic variations related to the metabolism of each vitamin are analysed. Possible deficiencies are determined so that our specialists are able to adapt your diet to improve your health and prevent putative diseases related to the lack of vitamins.

VITAMINS	DESCRIPTION	RESULTS
Vitamin A	Normal vitamin A metabolism. Ensure daily recommended intake.	
Vitamin B⁶	High risk of vitamin B6 deficiency. Increase daily vitamin B6 intake. Supplementation should be evaluated.	
Vitamin B⁹	Low risk of folate deficiency. Ensure daily recommended intake.	
Vitamin B¹²	Low risk of vitamin B12 deficiency. Ensure daily recommended intake.	
Vitamin C	Normal vitamin C metabolism and levels. Ensure daily recommended intake.	
Vitamin D	Low risk of Viamin D deficiency. Ensure daily recommended intake.	
Vitamin E	High risk of Vitamin E deficiency. Ensure daily recommended intake. Supplementation strategies would be recommended.	

INDICATIONS

■ Positive effect

■ Medium-positive effect

■ Medium-negative effect

■ Negative effect

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Genetic results overview

3

3.14.

CATEGORY	DESCRIPTION
Ca Mg Fe Se Minerals	Please find below the different analysed categories related to food supplementation needs. • Calcium malabsorption risk LOW RISK OF CALCIUM MALABSORPTION Pg. 74 • Predisposition to dysregulated calcium levels SLIGHTLY INCREASED RISK OF DYSREGULATED PLASMA CALCIUM LEVELS Pg. 75 • Risk of iron overload LOW RISK OF HEMOCHROMATOSIS Pg. 76 • Risk of low iron plasma levels LOW RISK OF DECREASED IRON LEVELS Pg. 77 • Predisposition to dysregulated magnesium levels MEDIUM-HIGH RISK OF DYSREGULATED MAGNESIUM LEVELS Pg. 78 • Predisposition to dysregulated selenium levels NO ADDITIONAL RISK OF DYSREGULATED SELENIUM LEVELS Pg. 79 • Sodium sensitivity MEDIUM-LOW SODIUM SENSITIVITY Pg. 80

3.15.

CATEGORY	DESCRIPTION																																
Effectiveness of diets	13 genetic variations related to the metabolism of various nutrients are analyzed in this section. This information allow us to develop a personalized plan aimed at improving your eating habits and exercise, that will help you achieve your weight goals, improve your muscle and bone mass, lower the fat mass and maintain a balanced and healthy diet. Efficacy of low carbohydrate diets <table><tr><th>MARKER</th><th>RESULTS</th></tr><tr><td>KCTD10</td><td></td></tr><tr><td>MMAB</td><td></td></tr></table> 50.16% Efficacy of low fat diets <table><tr><th>MARKER</th><th>RESULTS</th></tr><tr><td>PPARG</td><td></td></tr><tr><td>GHSR</td><td></td></tr><tr><td>APOA2</td><td></td></tr><tr><td>SH2B1</td><td></td></tr><tr><td>TCF7L2</td><td></td></tr><tr><td>FTO</td><td></td></tr></table> 39.51% Efficacy of low calorie diets <table><tr><th>MARKER</th><th>RESULTS</th></tr><tr><td>PPARG</td><td></td></tr><tr><td>ADIPOQ</td><td></td></tr><tr><td>LEPR</td><td></td></tr><tr><td>ACSL5</td><td></td></tr><tr><td>ADRB2</td><td></td></tr></table> 9.11%	MARKER	RESULTS	KCTD10		MMAB		MARKER	RESULTS	PPARG		GHSR		APOA2		SH2B1		TCF7L2		FTO		MARKER	RESULTS	PPARG		ADIPOQ		LEPR		ACSL5		ADRB2	
MARKER	RESULTS																																
KCTD10																																	
MMAB																																	
MARKER	RESULTS																																
PPARG																																	
GHSR																																	
APOA2																																	
SH2B1																																	
TCF7L2																																	
FTO																																	
MARKER	RESULTS																																
PPARG																																	
ADIPOQ																																	
LEPR																																	
ACSL5																																	
ADRB2																																	

INDICATIONS

Positive effect

Medium-positive effect

Medium-negative effect

Negative effect

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

4

Recommended nutritional plan

CGATCGATCGATCGATCGATCGTATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTG
GACCGACAGTCATCCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGATCGATCGATCGA
ATCGATCGATCGATCGATCGACGTACTGATCGATCGAATCGATCGATCGATCGACGTACTGATCGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGAT
ATCGATCGATCGATCGATCGATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTCGATCGATCGATCGAGTACTGAT

Recommended nutritional plan

4



The most effective diet for your patient, after the genetic analysis, would be

LOW IN CARBOHYDRATES
INTEGRATED NUTRITIONAL PLAN

Check the **FOOD LIST** recommended for you

• Patient Name: **Maria Salina Cooper**

• Gender: **Female**

• Date of Birth: **03-06-1974**

• Sample code: **NUT04947AA**

• Sample date: **04-06-2025**

• Date of the results: **04-06-2026**

Daily food intake

• Allowed adjusting the amounts and/or frequency*

• Allowed without raising the recommended quantities and/or frequency*

• Reduce the amount and/or frequency

* Observations on recommended foods are a suggestion based on the genetic test results. They should be evaluated by a professional and accurately adapted to the clinical history, lifestyle, habits, exercise, medication and psychological status.

On the food table, we have incorporated specific symbols for the recommendations or vitamin deficiencies based on the data included in the test. When several foods from a category have a similar level of recommendation, those whether they will have a positive effect or negative impact in the diet plan. Find out more.

Recommendable

Avoid consumption

Caffeine intolerance

Fructose intolerance

Gluten intolerance

Lactose intolerance

Alcohol

Carbohydrate

Lipid

Fat

Asthaxanthin intolerance

Carrot intolerance

Egg intolerance

Figs intolerance

Galactose intolerance

Ginger intolerance

Tomato intolerance

Monounsaturated Fatty Acids (MUFAs)

Polysaturated Fatty Acids (PUFAs)

Starch

Glucose

Salt

Kiwi intolerance

Nuts intolerance

Papaya intolerance

Pineapples intolerance

Cow-milk protein intolerance

Sea food intolerance

Soya intolerance

Fagron NutriGen

Food list

www.fagrongenomics.com

Fagron
genomics

• Patient Name: **Maria Salina Cooper**

• Gender: **Female**

• Date of Birth: **03-06-1974**

• Sample code: **NUT04947AA**

• Sample date: **04-06-2025**

• Date of the results: **04-06-2026**

Fruits and derivatives

FOOD NUTRIENT CONTENT

• Allowed adjusting the amounts and/or frequency*

• Allowed without raising the recommended quantities and/or frequency*

• Reduce the amount and/or frequency

• Restrict, occasionally! in small quantities*

NutriGen Report - Food list

Reviewed by: (Lab technician) - (09-06-2026)

Validated by: (Lab responsible) - (15-06-2026)

4 / 22

Fagron
genomics

NutriGen Report - Food list

Reviewed by: (Lab technician) - (09-06-2026)

Validated by: (Lab responsible) - (15-06-2026)

7 / 22

Fagron
genomics

NutriGen Report - Doctor
Reviewed by: (Lab technician) - (09-06-2026)
Validated by: (Lab responsible) - (15-06-2026)

17 / 87

Fagron
genomics

5

Recommended supplements

CGATCGATCGATCGATCGATCGTATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTG
GACCGACAGTCATCCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGATCGATCGATCGA
ATCGATCGATCGATCGATCGACGTACTGATCGATCGAATCGATCGATCGATCGACGTACTGATCGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGAT
ATCGATCGATCGATCGATCGATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTCGATCGATCGATCGAGTACTGAT

Recommended supplements

5

The supplements recommended to combat overweight and ageing are divided into 3 phases



DETOX

DETOX I
Detoxification
(oxidation)
30 days

- Glutathione (Reduced glutathione)
- Silymarin
- Quercetin
- Selenium (Selenium yeast)
- Resveratrol
- Nicotinamide (Niacinamida)
- Astaxanthin
- Vitamin B2 (Riboflavine)
- GreenSelectTM (Green tea extract)

DETOX II
Detoxification
(conjugation)
30 days

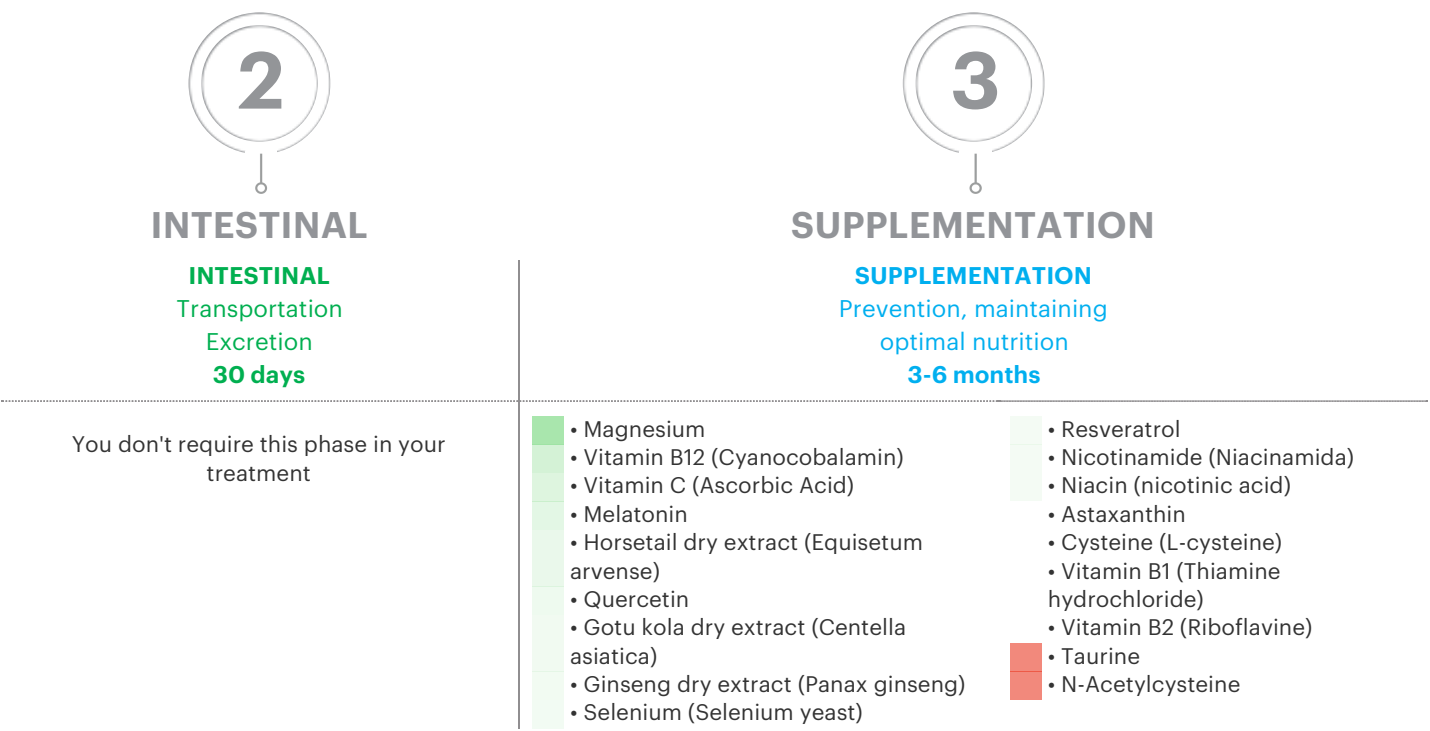
- Magnesium
- Vitamin B12 (Cyanocobalamin)
- Glutathione (Reduced glutathione)
- Silymarin
- Quercetin
- Gotu kola dry extract (Centella asiatica)
- Resveratrol
- Taurine
- N-Acetylcysteine

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Recommended supplements

5

The supplements recommended to combat overweight and ageing are divided into 3 phases



* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Recommended formulations

Recommended formulations

6



Detoxification
(oxidation)
30 days

Suggested formula: **CAPSULE**

Detox 1 capsule

Glutathione (Reduced glutathione)	200 mg
Selenium (Selenium yeast)	75 mcg
Resveratrol	91 mg

Daily dosage

Treatment for 30 days in accordance with the physician's decision.

Signature of the prescribing physician:

Dr

Physician Registration

Date of prescription

Address:

Signature:

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Recommended formulations

6



Detoxification
(conjugation)
30 days

You don't require this phase in your treatment.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Recommended formulations

6



Transportation
Excretion
30 days

You don't require this phase in your treatment.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Recommended formulations

6



Prevention, maintaining
optimal nutrition
3-6 months

Suggested formula: **CAPSULE**

Supplementation capsule

Magnesium	350 mg
Vitamin B12 (Cyanocobalamin)	550 mcg
Vitamin C (Ascorbic Acid)	200 mg
Melatonin	2 mg
Horsetail dry extract (Equisetum arvense)	150 mg
Quercetin	100 mg

Daily dosage

Treatment for 3-6 months in accordance with the physician's decision.

Signature of the prescribing physician:

Dr

Physician Registration

Date of prescription

Address:

Signature:

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Annex to the formulation

6

Composites daily dosage recommendation

Composite Name	Min	Max	Unit	Composite Name	Min	Max	Unit
Astaxanthin	4	18	mg	Quercetin	250	1000	mg
Cysteine (L-cysteine)	100	500	mg	Resveratrol	50	500	mg
Ginseng dry extract (Panax ginseng)	100	200	mg	Selenium (Selenium yeast)	50	200	mcg
Glutathione (Reduced glutathione)	200	3000	mg	Silymarin	80	200	mg
Gotu kola dry extract (Centella asiatica)	50	750	mg	Vitamin B1 (Thiamine hydrochloride)	2	100	mg
Horsetail dry extract (Equisetum arvense)	100	900	mg	Vitamin B12 (Cyanocobalamin)	250	1000	mcg
Magnesium	250	400	mg	Vitamin B2 (Riboflavine)	1	30	mg
Niacin (nicotinic acid)	1	50	mg	Vitamin C (Ascorbic Acid)	80	1000	mg
Nicotinamide (Niacinamida)	1	500	mg				

7

Complete genetic results

- 7.1. Morphological genetics for weight control

7.1.1. Genetic risk of overweight/obesity

7.1.2. Risk of rebound weight gain

7.1.3. Risk of increased BMI

7.1.4. Basal metabolic rate (burn calories at rest)

7.1.5. Weight loss capability during diet interventions
- 7.2. Behavioural genetics in food intake

7.2.1. Appetite and anxiety risk

7.2.2. Satiety: Feeling Full
- 7.3. Flavour sensitivities

7.3.1. Bitter taste sensitivity

7.3.2. Salt sensitivity

7.3.3. Sweet flavour preference
- 7.4. Fat metabolism

7.4.1. Response to monounsaturated fats (MUFAs)

7.4.2. Response to polyunsaturated fats (PUFAs)

7.4.3. Response to fat intake to improve the HDL levels
- 7.5. Lipid metabolism

7.5.1. Predisposition to reduced HDL levels

7.5.2. Predisposition to increased levels of triglycerides

7.5.3. Predisposition to increased oxidation of LDL

7.5.4. Risk of increased cholesterol LDL levels

7.5.5. Risk of unbalanced Triglycerides/HDL ratio
- 7.6. Carbohydrate metabolism

7.6.1. Capability to digest starchy food

7.6.2. Refined carbohydrate sensitivity

7.6.3. Carbohydrates and HDL levels predisposition

7.6.4. Carbohydrates and LDL levels
- 7.7. Glucose metabolism

7.7.1. Risk of increased glucose levels in plasma after fasting

7.7.2. Risk of insulin resistance

7.7.3. Risk of Type-II diabetes
- 7.8. Efficacy of exercise

7.8.1. Benefits from endurance exercise for improving HDL levels

7.8.2. Exercise to reduce body fat
- 7.9. Detoxification imbalances

7.9.1. Antioxidant capability
- 7.10. Intolerance

7.10.1. Lactose intolerance risk

7.10.2. Risk of celiac disease

7.10.3. Fructose intolerance risk

7.10.4. Caffeine metabolism

7.10.5. Alcohol metabolism
- 7.11. Hormones

7.11.1. Leptin

7.11.2. Visfatin

7.11.3. Ghrelin

7.11.4. Adiponectin
- 7.12. Inflammation

7.12.1. TNF-α

7.12.2. IL-6

7.12.3. IL-10
- 7.13. Vitamins

7.13.1. Vitamin A

7.13.2. Vitamin B6

7.13.3. Vitamin B9 (folate)

7.13.4. Vitamin B12

7.13.5. Vitamin C

7.13.6. Vitamin D

7.13.7. Vitamin E
- 7.14. Minerals

7.14.1. Calcium malabsorption risk

7.14.2. Predisposition to dysregulated calcium levels

7.14.3. Risk of iron overload

7.14.4. Risk of low iron plasma levels

7.14.5. Predisposition to dysregulated magnesium levels

7.14.6. Predisposition to dysregulated selenium levels

7.14.7. Sodium sensitivity
- 7.15. Effectiveness of diets

7.15.1. Efficacy of low calorie diets

7.15.2. Efficacy of low carbohydrate diets

7.15.3. Efficacy of low fat diets

CGATCGATCGATCGATCGATCGGTATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTG
GACCGA^CCAGTCATCCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGATCGATCGATCGA
ATCGATCGATCGATCGATCGACGTACTGATCGA^ATTCGATCGAATCGATCGATCGATCGACGTACTGATCGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGATCGATCGATCGAGTACTGAT
ATCGATCGATCGATCGATCGATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTACTGA
ATCGACGTACTGATCGATCGATCGAATCGATCGATCGATCGACGTAC^CGATCGATCGATCGAGTACTGAT

Complete genetic results

7

7. 1 Morphological genetics for weight control



7. 1. 1. Genetic risk of overweight/obesity

RESULT		ABOUT		
MEDIUM-HIGH RISK OF OVERWEIGHT/OBESITY		Key genetic predisposition genes to obesity and weight gain are analysed. Obesity is influenced by the interplay between external factors (such as diet and/or physical activity) and is highly linked to individual genetics. Genetics highly determine how the body processes or metabolizes fats and/or nutrients. Therefore, understanding our own genetics is important to control obesity and as a key weight reduction tool.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
MC4R	rs2229616	CC	HIGH	Higher risk of obesity. High predisposition to increased glycosylated hemoglobin (increased risk of type 2 diabetes) and decreased HDL-cholesterol levels.
SH2B1	rs7498665	AG	MEDIUM	Increased predisposition to obesity and insulin resistance.
FTO	rs9939609	TA	MEDIUM	Predisposition to obesity, related to insulin resistance, hyperphagia, and increased risk of type 2 diabetes.
FTO	rs1121980	GA	MEDIUM	Increased risk of obesity related with insulin resistance, hyperphagia, and increased risk of type 2 diabetes.
MC4R	rs17700633	GA	MEDIUM	Increased risk of obesity and type 2 diabetes.

INDICATIONS

LOW RISK OF OVERWEIGHT/OBESITY

Reduced risk of obesity due to inherited genetic factors.

MEDIUM-LOW RISK OF OVERWEIGHT/OBESITY

Medium-low risk of obesity due to inherited genetic factors.

MEDIUM-HIGH RISK OF OVERWEIGHT/OBESITY

Medium-high risk of obesity due to inherited genetic factors. Other factors such as intake due to anxiety or low satiety may explain excess weight.

HIGH RISK OF OVERWEIGHT/OBESITY

High risk of obesity due to inherited genetic factors. Other factors such as intake due to anxiety or low satiety may explain excess weight.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 1 Morphological genetics for weight control



7. 1. 2. Risk of rebound weight gain

RESULT		ABOUT		
HIGH REBOUND EFFECT		Individuals with certain genetic variants of the ADIPOQ gene were found to be more susceptible to regain weight after weight loss interventions (rebound effect).		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ADIPOQ	rs17300539	GG	HIGH	Predisposition to regain weight after dieting.
INDICATIONS				
LOW REBOUND EFFECT Low risk of rebound weight after diet interventions. Normal weight loss capacity.	MEDIUM-LOW REBOUND EFFECT Medium-low risk of rebound weight after diet interventions. Normal weight loss capacity.	MEDIUM-HIGH REBOUND EFFECT Medium-high risk of rebound weight after diet interventions. Lower weight loss capability than normal during interventions.	HIGH REBOUND EFFECT High risk of rebound weight after diet interventions. Lower weight loss capability than normal during interventions. It will require an extra effort to loose weight and keep it off afterwards.	

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 1 Morphological genetics for weight control



7. 1. 3. Risk of increased BMI

RESULT		ABOUT		
MEDIUM-LOW RISK OF INCREASED BMI		The predisposition to increase waist circumference and body mass index (BMI) is analyzed. BMI is used to determine whether an individual is in a healthy weight range for the correspondent height. It is useful to consider BMI alongside waist circumference, as waist measurement helps to assess risk by measuring the amount of fat carried around the middle.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
MC4R	rs12970134	GA	MEDIUM	Increased risk of increased BMI, increased waist circumference, and insulin resistance.
MC4R	rs17782313	TC	MEDIUM	Increased risk of increased BMI, increased waist circumference, and insulin resistance.
SH2B1	rs4788102	GA	MEDIUM	Increased risk of increased BMI, increased waist circumference, and type 2 diabetes.

INDICATIONS

LOW RISK OF INCREASED BMI
Reduced risk of increased BMI, waist circumference and insulin resistance due to genetics.

MEDIUM-LOW RISK OF INCREASED BMI
Medium-low risk of increased BMI, waist circumference and insulin resistance due to genetics.

MEDIUM-HIGH RISK OF INCREASED BMI
Medium-high risk of increased BMI, waist circumference and insulin resistance due to genetics.

HIGH RISK OF INCREASED BMI
High risk of increased BMI, waist circumference and insulin resistance due to genetics.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 1 Morphological genetics for weight control



7. 1. 4. Basal metabolic rate (burn calories at rest)

RESULT		ABOUT		
HIGH BURNER AT REST		The predisposition to an increase/decrease in energy expenditure while resting is analysed. Some people have a higher tendency then others to expend less energy when not performing any physical activity, which supports weight gain.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
FABP2	rs1799883	CC	LOW	Predisposition to normal resting metabolic rate.
LEPR	rs2025804	AA	LOW	Predisposition to normal resting metabolic rate.

INDICATIONS

HIGH BURNER AT REST
High energy/calorie burning capacity at rest.

MEDIUM-HIGH BURNER AT REST
Medium-high capacity to burn energy/calories at rest.

MEDIUM-LOW BURNER AT REST
Medium-low capacity of energy/calorie burning at rest.

LOW BURNER AT REST
Low energy/calorie burning capacity at rest.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 1 Morphological genetics for weight control



7. 1. 5. Weight loss capability during diet interventions

RESULT		ABOUT		
SLOW WEIGHT LOSS		The predisposition to an increase/decrease in weight loss during diet interventions is analysed. Some people have a higher tendency than others to lose weight when they follow a diet intervention. Lower capabilities will imply a longer time to accomplish the goals and would require a stricter intervention.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ACSL5	rs2419621	CC	HIGH	Predisposition to slow diet-induced weight loss.

INDICATIONS

RAPID WEIGHT LOSS

Diet interventions should be successful due to a higher capability to reduce weight while on diet.

NORMAL WEIGHT LOSS

Diet interventions should be successful due to a normal capability to reduce weight while on diet. However it may take a minimum of 3-6 months to be effective.

SLIGHTLY SLOW WEIGHT LOSS

Standard diet interventions could not be successful due to a low capability to reduce weight while on diet. Specialized treatments would be recommended.

SLOW WEIGHT LOSS

Diet interventions should contain a complete approach for the patient, both nutritional and psychological, due to the lower capability to reduce weight while on diet. Specialised treatments will be recommended.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 2 Behavioural genetics in food intake



7. 2. 1. Appetite and anxiety risk

●

INCREASED APPETITE AND ANXIETY RISK

RESULT		ABOUT		
		Genetic variations affecting appetite and anxiety related to eating are analysed. Appetite is a phenomenon created by our nervous system which results in a desire to eat, either by necessity or by pleasure, and in which external factors (such as odors, flavours, appearance and presentation of food) are involved. It has been seen in numerous studies that the appetite or desire to eat can also have genetic causes that can determine inhibition of intake or reduced feeling of being full. Anxiety related to food intake can be caused by periods of stress, but it has also been seen that there is an important genetic component that makes us more prone to anxiety and translates into compulsive eating more easily. The main parameters related to genetic predisposition to deregulated levels of appetite and anxiety in food intake, increased risk of obesity, increased food intake and reduced fullness are analysed below. Knowing how these genetic processes affect your diet allows proper handling of meals.		

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
COMT	rs4680	GG	HIGH	High risk of overeating and bulimia nervosa associated with an increased risk of type 2 diabetes and impaired glucose tolerance.
NMB	rs1051168	GG	LOW	Normal risk of eating disinhibition.
DRD2	rs1800497	GG	LOW	No predisposition to emotional eating.
MC4R	rs2229616	CC	HIGH	Predisposition to binge eating.
DRD2	rs6277	AA	HIGH	Predisposition to binge eating.

INDICATIONS

●

NORMAL APPETITTE AND ANXIETY RISK

Normal or well-balanced regulation of appetite and eating-related anxiety.

●

SLIGHTLY INCREASED APPETITE AND ANXIETY RISK

Medium-low dysregulation of the appetite, leading to some levels of anxiety affecting food intake.

●

INCREASED APPETITE AND ANXIETY RISK

Medium-high dysregulation of the appetite, leading to elevated levels of anxiety affecting food intake. Appetite suppressants may be helpful.

●

HIGHLY INCREASED APPETITE AND ANXIETY RISK

High dysregulation of the appetite, leading to high levels of anxiety affecting food intake. Appetite suppressants may be required and possibly anxiolytic prescription upon medical decision.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 2 Behavioural genetics in food intake



7. 2. 2. Satiety: Feeling Full

RESULT		ABOUT		
SLIGHTLY LOWER SATIETY		The perception of feeling full and satisfied after food intake is different within individuals. This is particularly important as the longer it takes to reach this feeling, the more food intake will occur, contributing to weight gain.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
FTO	rs9939609	TA	MEDIUM	Slight predisposition to diminished satiety. Increased risk of obesity.

INDICATIONS

NORMAL SATIETY

Normal perception of satiety after eating, activated after 15-20 minutes of the start of the meal.

SLIGHTLY LOWER SATIETY

Slightly reduced perception of satiety after eating a meal. Try to eat slower to allow the satiety center to be activated.

LOWER SATIETY

Reduced perception of satiety after eating a meal. Eat slower to allow the satiety center to be activated.

VERY LOW SATIETY

Very low perception of satiety after eating a meal. Eat very slow to allow the satiety center to be activated. Incorporate satiating food in your daily diet.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 3 Flavour sensitivities



7. 3. 1. Bitter taste sensitivity

RESULT		ABOUT		
NORMAL BITTER TASTE SENSITIVITY		Sensitivity to bitter flavours is deeply linked to genetics. A high sensitivity to bitter flavours is usually linked to increased salt consumption. Therefore there is a higher predisposition to cardiovascular risks when extra salt is consumed intending to mask the bitter flavours.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
TAS2R38	rs1726866	GA	LOW	Predispositon to normal sensitivity to bitter taste.
TAS2R38	rs713598	CG	LOW	Predispositon to normal sensitivity to bitter taste.

INDICATIONS

NORMAL BITTER TASTE SENSITIVITY

Normal or decreased sensitivity to bitter flavours. No extra salt should be consumed for this reason.

SLIGHTLY INCREASED BITTER TASTE SENSITIVITY

Slightly increased sensitivity to bitter flavours. No extra salt should be consumed for this reason.

INCREASED BITTER TASTE SENSITIVITY

Increased sensitivity to bitter flavours. Try to minimize bitter-tasting food, since it may lead to an increased consumption of salt.

HIGHLY INCREASED TASTE SENSITIVITY

High sensitivity to bitter flavours. Try to avoid bitter-tasting food, since it may lead to an increased consumption of salt.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7. 3 Flavour sensitivities



7. 3. 2. Salt sensitivity

●

MEDIUM-LOW SALT SENSITIVITY

RESULT		ABOUT		
		Salt sensitivity is defined as a physiological trait by which blood pressure shows changes parallel to changes in salt intake. In many individuals, when salt intake increases, the excess amount is excreted by the way of kidney or sweat. However, there are some individuals where this mechanism is faulty and increased salt is retained and manifests as high blood pressure.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ACE	rs4343	GA	MEDIUM	Predisposition to increased salt sensitivity associated with increased risk of salt sensitive hypertension.

INDICATIONS

●

LOW SALT SENSITIVITY

Normal salt sensitivity: no increased blood pressure risk due to salt consumption.

●

MEDIUM-LOW SALT SENSITIVITY

Slightly increased salt sensitivity: moderately increased blood pressure risk due to salt consumption.

●

MEDIUM-HIGH SALT SENSITIVITY

Increased salt sensitivity: increased blood pressure risk due to salt consumption. Reduce current salt consumption, if daily intake is high.

●

HIGH SALT SENSITIVITY

High salt sensitivity: high blood pressure risk due to salt consumption. Reduce current salt consumption, if daily intake is high.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 3 Flavour sensitivities



7. 3. 3. Sweet flavour preference

RESULT		ABOUT		
HIGHLY INCREASED SWEET PREFERENCE		Increased desire to eat sweet food due to an incapacity of tasting sweet flavours.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
SLC2A2	rs5400	GA	HIGH	Predisposition to increased preference for sugar-containing foods.

INDICATIONS

NORMAL SWEET PREFERENCE

Normal taste of sweet flavour. No excess sugar intake should be required.

SLIGHTLY INCREASED SWEET PREFERENCE

Slight incapacity to taste sweet flavours. This will lead to an increase in sugar consumption and obesity risk.

INCREASED SWEET PREFERENCE

Incapacity to taste sweet flavours. This will lead to an increase in the sugar consumption and obesity risk. Consider using artificial sweeteners in your diet.

HIGHLY INCREASED SWEET PREFERENCE

Major incapacity to taste sweet flavours. This will lead to an increase in the sugar consumption and obesity risk. Consider using artificial sweeteners in your diet.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 4 Fat metabolism



7. 4. 1. Response to monounsaturated fats (MUFAs)

RESULT		ABOUT		
VERY LOW MUFA METABOLISM		The predisposition to a higher/lower capacity to metabolize monounsaturated fatty acids (MUFAs) is analysed. MUFAs are a class of fatty acids found in foods such as olive oil, nuts and avocados. The beneficial effects of MUFAs on cardiovascular disease risk and blood lipid profiles have been extensively studied: dietary MUFAs decrease oxidized LDL, LDL cholesterol, total cholesterol, and triglyceride concentrations, without the concomitant decrease in HDL typically seen with low-fat diets.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ADIPOQ	rs17300539	GG	HIGH	No predisposition to reduce BMI and decrease obesity risk in response to monounsaturated fatty acids (MUFA) intake.

INDICATIONS

FAST MUFA METABOLISM Normal capability of burning monounsaturated fat (MUFA). Increased capability to intake and metabolize MUFA with low weight gain.	MEDIUM MUFA METABOLISM Medium capability of burning monounsaturated fat (MUFA). MUFA intake may lead to low weight gain unless a high-fat diet is followed.	LOW MUFA METABOLISM Low capability of burning monounsaturated fat (MUFA). Direct correlation of high-MUFA intake and weight gain due to fat accumulation.	VERY LOW MUFA METABOLISM Very low capability of burning monounsaturated fat (MUFA). Direct correlation on high-MUFA intake and weight gain due to fat accumulation.
--	---	---	---

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 4 Fat metabolism



7. 4. 2. Response to polyunsaturated fats (PUFAs)

RESULT		ABOUT		
LOW PUFA METABOLISM		The predisposition to a higher/lower capacity to metabolize polyunsaturated fatty acids (PUFA) and to improve the lipidic profile (decreased LDL-levels) with PUFA intake is analysed. Polyunsaturated fatty acids are necessary to build cell membranes and nerve coverings as well as for proper blood clotting, muscle movement and inflammation. There are two main types of polyunsaturated fats: omega-3 fatty acids and omega-6 fatty acids. Both types provide health benefits.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARG	rs1801282	CC	LOW	Predisposition to improve lipid profile (LDL and total cholesterols) and reduce BMI in response to a PUFA-rich diet.
FADS1	rs174547	CC	HIGH	Age-related predisposition to reduced PUFA biosynthetic capacity and low plasma omega 3 concentration.

INDICATIONS

FAST PUFA METABOLISM Normal capability of burning polyunsaturated fat (PUFA). Increased capability to intake and metabolize PUFA with low weight gain. Improved lipidic profiles with PUFA intake.	MEDIUM PUFA METABOLISM Medium capability of burning polyunsaturated fat (PUFA). PUFA intake may lead to low weight gain unless a high-fat diet is followed. Improved lipidic profiles with PUFA intake.	LOW PUFA METABOLISM Low capability of burning polyunsaturated fat (PUFA). Direct correlation of high-PUFA intake and weight gain due to fat accumulation.	VERY LOW PUFA METABOLISM Very low capability of burning polyunsaturated fat (PUFA). Direct correlation of high-PUFA intake and weight gain due to fat accumulation.
--	---	---	---

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 4 Fat metabolism



7. 4. 3. Response to fat intake to improve the HDL levels

RESULT		ABOUT		
VERY LOW BENEFITS TO IMPROVE HDL		The predisposition to have increased or reduced levels of HDL is analyzed according to the genetic situation of liver lipases. With this category, we understand if a low fat diet is a good strategy to regulate cholesterol levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
LIPC	rs1800588	CC	HIGH	No predisposition to improve HDL cholesterol levels in response to low fat diet.

INDICATIONS

HIGH BENEFITS TO IMPROVE HDL

A low fat diet will be of great help in increasing HDL levels.

MEDIUM-HIGH BENEFITS TO IMPROVE HDL

A low fat diet will be a good support to increase HDL levels.

MEDIUM-LOW BENEFITS TO IMPROVE HDL

Low fat diet will not be enough to increase HDL levels.

VERY LOW BENEFITS TO IMPROVE HDL

Low fat diet will not be enough to increase HDL levels.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 5 Lipid metabolism



7. 5. 1. Predisposition to reduced HDL levels

RESULT		ABOUT		
NORMAL HDL LEVELS		Although environmental factors play a role, variation in HDL levels are at least 50% genetically determined. In this category the main genes involved in the predisposition to higher or lower HDL levels are analysed.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
APOA5	rs662799	AA	LOW	Predisposition to normal levels of HDL cholesterol.
CETP	rs5883	CT	LOW	Predisposition to increased HDL cholesterol levels.

INDICATIONS

NORMAL HDL LEVELS

Normal regulation of HDL levels. No increased risk of cardiovascular risk.

SLIGHTLY DECREASED HDL LEVELS

Slightly lower HDL levels leading to increased cardiovascular risk.

REDUCED HDL LEVELS

Lower HDL levels leading to increased cardiovascular risk.

HIGLY REDUCED HDL LEVELS

Very low HDL levels leading to increased cardiovascular risk.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 5 Lipid metabolism



7. 5. 2. Predisposition to increased levels of triglycerides

RESULT		ABOUT		
HIGHLY INCREASED TRIGLYCERIDES		Triglycerides are a type of fat (lipid) found in your blood. When you eat, your body converts any calories it doesn't need to use right away into triglycerides. The triglycerides are stored in your fat cells. Later, hormones release triglycerides for energy between meals. If you regularly eat more calories than you burn, particularly from high-carbohydrate foods, you may have high triglycerides (hypertriglyceridemia). In this category we analyse the genes related to the predisposition of having increased levels of triglycerides.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARG	rs1801282	CC	HIGH	Predisposition to increased levels of triglycerides.

INDICATIONS

TRIGLYCERIDES NOT INCREASED No predisposition to increased triglyceride levels.	SLIGHTLY INCREASED TRIGLYCERIDES Slight predisposition to increased triglyceride levels.	INCREASED TRIGLYCERIDES Medium-high predisposition to increased triglyceride levels.	HIGHLY INCREASED TRIGLYCERIDES High predisposition to increased triglyceride levels
--	---	---	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 5 Lipid metabolism



7. 5. 3. Predisposition to increased oxidation of LDL

RESULT		ABOUT		
SLIGHTLY INCREASED LDL OXIDATION		Oxidized low-density lipoprotein (LDL) is a harmful type of cholesterol that is produced in your body when normal LDL cholesterol is damaged by chemical interactions with free radicals. These, and a related series of inflammatory responses can result in atherosclerosis, which is the hardening of the arteries. The resulting decrease in blood flow in your arteries increases your chances of having a heart attack or a stroke. You can produce high levels of oxidized LDL if you have excessive free radical formation or simply high LDL cholesterol levels. In this category, the genes related to an increased predisposition to oxidize LDL are analysed.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
APOB	rs676210	GA	MEDIUM	Predisposition to increased LDL oxidation.

INDICATIONS

NOT INCREASED LDL OXIDATION Normal LDL oxidation.	SLIGHTLY INCREASED LDL OXIDATION Moderate increase in the LDL oxidation. Increased risk of atherosclerosis.	INCREASED LDL OXIDATION Increased LDL oxidation. Increased risk of atherosclerosis. Strategies for reducing LDL levels would be recommended.	HIGHLY INCREASED LDL OXIDATION Higly increased LDL oxidation and risk of atherosclerosis. Intense strategies for reducing LDL levels should be initiated.
--	--	---	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 5 Lipid metabolism



7. 5. 4. Risk of increased cholesterol LDL levels

RESULT		ABOUT		
INCREASED LDL LEVELS		Low-density lipoprotein (LDL) is one of the five major groups of lipoprotein which transport all fat molecules around the body in extracellular water. LDL delivers fat molecules to cells. LDL can contribute to atherosclerosis if it is oxidized within the walls of arteries. In this category, the genes related to the risk of having increased cholesterol LDL levels in your body are analysed.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
CELSR2	rs12740374	GG	HIGH	No predisposition to lower LDL cholesterol levels.
HNF1A	rs2650000	CC	LOW	Predisposition to normal LDL cholesterol levels.
LDLR	rs6511720	GG	HIGH	High risk of increased LDL cholesterol levels.
ABCG8	rs6544713	TC	MEDIUM	Predisposition to slightly increased LDL cholesterol levels.

INDICATIONS

NOT INCREASED LDL LEVELS

Lower risk of high LDL levels

SLIGHTLY INCREASED LDL LEVELS

Moderate risk of high LDL levels

INCREASED LDL LEVELS

High risk of high LDL levels.

HIGHLY INCREASED LDL LEVELS

Very high risk of high LDL levels.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 5 Lipid metabolism



7. 5. 5. Risk of unbalanced Triglycerides/HDL ratio

RESULT		ABOUT		
SLIGHTLY INCREASED TG/HDL RATIO		The predisposition to an unbalanced Triglyceride/HDL cholesterol (TG/HDL-C) ratio is analysed. High TG/HDL ratio has been identified as a risk factor for cardiovascular (CV) diseases.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
HMGCR	rs3846663	CT	MEDIUM	Predisposition to slightly higher triglyceride (TG) levels, and increased TG/HDL cholesterol ratio.

INDICATIONS

NORMAL TG/HDL RATIO
Not associated with increased TG/HDL ratio.

SLIGHTLY INCREASED TG/HDL RATIO
Slightly associated with increased TG/HDL ratio.

INCREASED TG/HDL RATIO
Increased TG/HDL ratio leads to a highly increased risk of cardiovascular pathologies. Risk of insulin insensitivity.

HIGHLY INCREASED TG/HDL RATIO
A very high TG/HDL ratio leads to a highly increased risk of cardiovascular pathologies. Risk of insulin insensitivity.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 6 Carbohydrate metabolism



7. 6. 1. Capability to digest starchy food

RESULT		ABOUT		
● INCREASED STARCH DIGESTION		The capability to break down starch from food is analysed. Amylase is an enzyme that catalyzes the hydrolysis of starch into sugars. Amylase is present in the saliva of humans and some other mammals, where it begins the chemical process of digestion. When starch is not properly processed, its consumption must be reduced in a diet plan.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
AMY1-AMY2	rs11577390	CT	MEDIUM	Predisposition to slightly increased expression of the amylase gene which is likely to enable more efficient starch digestion.
AMY1	rs4244372	TT	LOW	Predisposition to increased expression of the amylase gene which is likely to enable more efficient starch digestion.
INDICATIONS				
●	INCREASED STARCH DIGESTION Increased capability to digest starch from food due to an increase in the expression and the activity of amylase enzyme. It is known that reducing calories will be beneficial.			
●	MEDIUM STARCH DIGESTION Moderate capability to digest starch from food due to an increase in the expression and the activity of amylase enzyme. It is known that reducing calories will be beneficial.			
●	REDUCED STARCH DIGESTION Reduced capability to digest starch in food due to a decrease in amylase enzyme activity. It would be beneficial to decrease starch intake.			
●	HIGHLY REDUCED STARCH DIGESTION Highly reduced capability to digest starch in food due to a decrease in amylase enzyme activity. It would be beneficial to decrease starch intake.			

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 6 Carbohydrate metabolism



7. 6. 2. Refined carbohydrate sensitivity

RESULT		ABOUT		
VERY HIGH CARBOHYDRATE SENSITIVITY		Carbohydrate consumption initially produces a slight euphoria, followed by a sugar low, this is then replaced by tiredness. This adverse feeling causes a desire to snack more, perpetuating this unhealthy cycle, without ever feeling satisfied. In carbohydrates sensitive people the carbohydrate-insulin-serotonin connection has malfunctioned, or become desensitised and the amount of calories extracted by the consumption of refined carbohydrates is higher than average, also due to a continuous increase of its consumption.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
FABP2	rs1799883	CC	HIGH	Predisposition to increased sensitivity to refined carbohydrates.

INDICATIONS

NORMAL CARBOHYDRATE SENSITIVITY Normal calorie extraction from carbohydrate consumption.	MEDIUM CARBOHYDRATE SENSITIVITY Moderate calorie extraction from carbohydrate consumption. Medium risk of weight gain.	HIGH CARBOHYDRATE SENSITIVITY Increased calorie extraction from carbohydrate consumption. Higher risk of weight gain.	VERY HIGH CARBOHYDRATE SENSITIVITY Highly increased calorie extraction from carbohydrate consumption. Very high risk of weight gain.
---	---	--	---

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 6 Carbohydrate metabolism



7. 6. 3. Carbohydrates and HDL levels predisposition

RESULT		ABOUT		
HIGH RISK OF HDL DYSREGULATION		Carbohydrate intake has an implication on the regulation of cholesterol levels. We analyse the predisposition to increase or decrease the HDL cholesterol levels depending on carbohydrate intake.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
KCTD10	rs10850219	GG	HIGH	Predisposition to reduce HDL cholesterol levels in response to a carbohydrate-rich diet.

INDICATIONS

LOW RISK OF HDL DYSREGULATION

High carbohydrate consumption will not lead to a cholesterol dysregulation.

MEDIUM-LOW RISK OF HDL DYSREGULATION

High carbohydrate consumption may lead to slightly increased LDL and decreased HDL levels.

MEDIUM-HIGH RISK OF HDL DYSREGULATION

High carbohydrate consumption will lead to increased LDL and decreased HDL levels.

HIGH RISK OF HDL DYSREGULATION

High carbohydrate consumption will lead to highly increased LDL and decreased HDL levels.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 6 Carbohydrate metabolism



7. 6. 4. Carbohydrates and LDL levels

RESULT		ABOUT		
HIGH RISK OF LDL DYSREGULATION		Effect of carbohydrate intake in the regulation of cholesterol levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
MMAB	rs2241201	CG	HIGH	High risk to increase LDL-cholesterol levels and decrease HDL-cholesterol levels in response to high intake of carbohydrates.

INDICATIONS

LOW RISK OF LDL DYSREGULATION

High carbohydrate consumption will not lead to cholesterol dysregulation.

MEDIUM-LOW RISK OF LDL DYSREGULATION

High carbohydrate consumption will lead to very slight increased LDL and decreased HDL levels.

MEDIUM-HIGH RISK OF LDL DYSREGULATION

High carbohydrate consumption will lead to increased LDL and decreased HDL levels.

HIGH RISK OF LDL DYSREGULATION

High carbohydrate consumption will lead to highly increased LDL and decreased HDL levels.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 7 Glucose metabolism



7. 7. 1. Risk of increased glucose levels in plasma after fasting

RESULT		ABOUT		
MEDIUM-HIGH RISK OF HIGH GLUCOSE LEVELS		Fasting blood sugar levels give vital clues about how a person's body is managing blood sugar. Blood sugar tends to peak about an hour after eating and declines after that. High fasting blood sugar levels point to insulin resistance or diabetes. In this category, the genes related to the predisposition to an increased level of glucose after fasting are analysed, helping to understand how the body manages sugar.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PLIN1	rs2289487	CT	MEDIUM	Predisposition to slightly increased plasma glucose levels after fasting.
GHSR	rs490683	GG	HIGH	High risk of increased plasma glucose levels after fasting.

INDICATIONS

LOW RISK OF HIGH GLUCOSE LEVELS

Normal fasting plasma glucose levels. No increased risk of Type-II diabetes.

MEDIUM-LOW RISK OF HIGH GLUCOSE LEVELS

Normal or slightly increased fasting plasma glucose levels. No increased risk of Type-II diabetes.

MEDIUM-HIGH RISK OF HIGH GLUCOSE LEVELS

Increased fasting plasma glucose levels. Increased risk of Type-II diabetes.

HIGH RISK OF HIGH GLUCOSE LEVELS

High risk of Increased fasting plasma glucose levels. Increased risk of Type-II diabetes.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 7 Glucose metabolism



7. 7. 2. Risk of insulin resistance

MEDIUM-HIGH INSULIN RESISTANCE

Insulin resistance (also called metabolic syndrome) is when cells in your muscles, fat, and liver don't respond well to insulin and can't use glucose from your blood for energy. To make up for it, your pancreas makes more insulin. Over time, your blood sugar levels go up. Insulin resistance syndrome includes a group of problems like obesity, high blood pressure, high cholesterol, and Type-II diabetes. In this category the genetic predisposition towards a higher risk of insulin resistance is analysed.

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARG	rs1801282	CC	HIGH	High predisposition to insulin resistance.
ADIPOQ	rs17300539	GG	HIGH	High predisposition to insulin resistance.
TCF7L2	rs7903146	CT	MEDIUM	Increased predisposition to insulin resistance.
FTO	rs9939609	TA	MEDIUM	Increased predisposition to insulin resistance.
FTO	rs1121980	GA	MEDIUM	Increased predisposition to insulin resistance.

INDICATIONS

LOW INSULIN RESISTANCE
Low inherited risk of insulin resistance

MEDIUM-LOW INSULIN RESISTANCE
Medium-low inherited risk of insulin resistance

MEDIUM-HIGH INSULIN RESISTANCE
Medium-high inherited risk of insulin resistance

HIGH INSULIN RESISTANCE
High inherited risk of insulin resistance

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 7 Glucose metabolism



7. 7. 3. Risk of Type-II diabetes

●

MEDIUM-HIGH DIABETES TYPE-II RISK

Type-II diabetes mellitus (T2DM) is caused by complex interplay between multiple genetic and environmental factors. In this category, a complete analysis of the main genetic variants related to an increase in the risk of developing Type-II diabetes is analysed.

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARG	rs1801282	CC	HIGH	Increased risk of diabetes type 2.
PLIN1	rs2289487	CT	MEDIUM	Slightly increased risk of diabetes type 2.
TCF7L2	rs7903146	CT	MEDIUM	Increased risk of diabetes type 2.
FTO	rs9939609	TA	MEDIUM	Increased risk of diabetes type 2.
MC4R	rs17700633	GA	MEDIUM	Slight predisposition to obesity increasing the risk of type 2 diabetes.
CDKN2A/B	rs10811661	TT	HIGH	High risk of type 2 diabetes.
KCNQ1	rs2237892	CC	HIGH	Increased risk of type 2 diabetes.
CDKN2A, CDKN2B	rs2383208	AA	HIGH	High risk of type 2 diabetes.
CDKAL1	rs7756992	AA	LOW	Normal risk of type 2 diabetes.
TCF7L2	rs7901695	TC	MEDIUM	Slightly increased risk of type 2 diabetes.

●

LOW DIABETES TYPE-II RISK

Normal diabetes type-II risk.

●

MEDIUM-LOW DIABETES TYPE-II RISK

Medium-low risk of developing type-II diabetes.

●

MEDIUM-HIGH DIABETES TYPE-II RISK

Medium-high risk of developing type-II diabetes.

●

HIGH DIABETES TYPE-II RISK

High risk of developing type-II diabetes.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 8 Efficacy of exercise



7. 8. 1. Benefits from endurance exercise for improving HDL levels

RESULT		ABOUT		
VERY LOW BENEFITS FROM EXERCISE FOR IMPROVING HDL		The predisposition to improving the HDL cholesterol levels via exercising is analysed. The expected efficacy of exercise on cholesterol regulation differs between individuals and is highly dependant on your genetics.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARD	rs2016520	TT	HIGH	No predisposition to increase HDL cholesterol levels in response to endurance exercise.

INDICATIONS

HIGH BENEFITS FROM EXERCISE FOR IMPROVING HDL

Exercise will be strongly beneficial for cholesterol regulation (HDL increase).

MEDIUM-HIGH BENEFITS FROM EXERCISE FOR IMPROVING HDL

Exercise will be beneficial for cholesterol regulation (HDL increase).

MEDIUM-LOW BENEFITS FROM EXERCISE FOR IMPROVING HDL

Exercise alone will not be enough for cholesterol regulation.

VERY LOW BENEFITS FROM EXERCISE FOR IMPROVING HDL

Exercise alone will not be enough for cholesterol regulation.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 8 Efficacy of exercise



7. 8. 2. Exercise to reduce body fat

MEDIUM-HIGH BENEFIT FROM EXERCISE TO REDUCE FAT

ABOUT

The efficacy of physical activity to reduce body fat is different among all of us and the cause is mainly genetic. This is the reason why some people, even exercising daily tend to lose less weight than others exercising a couple of times a week. In this category, the genes related to the efficacy of exercise to reduce body fat are analysed.

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
FTO	rs9939609	TA	MEDIUM	Slight predisposition to lose fat during physical exercise.
FTO	rs1121980	GA	MEDIUM	Predisposition to lose fat slowly during physical exercise.
LIPC	rs1800588	CC	HIGH	No predisposition to benefit from physical exercise to increase HDL cholesterol levels.
LEP	rs7799039	GA	MEDIUM	Predisposition to medium response to exercise-induced fat loss.

INDICATIONS

HIGH BENEFIT FROM EXERCISE TO REDUCE FAT

An exercise strategy will be a very good option for weight loss. Exercise 3-4 times per week at medium-high intensity will be beneficial for slimming. Introduce also some diet restrictions.

MEDIUM-HIGH BENEFIT FROM EXERCISE TO REDUCE FAT

An exercise strategy may be a good option for weight loss. Exercise 2-3 times per week at medium-high intensity will be beneficial for slimming. Also introduce some diet restrictions.

MEDIUM-LOW BENEFIT FROM EXERCISE TO REDUCE FAT

An exercise strategy may not be the best option for weight loss. Rather introduce diet restrictions and institute healthy sport-related habits (walking, swimming at low intensity).

VERY LOW BENEFIT FROM EXERCISE TO REDUCE FAT

An exercise strategy may not be the best option for weight loss. Rather introduce diet restrictions and institute healthy sport-related habits (walking, swimming at low intensity).

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 9 Detoxification imbalances



7. 9. 1. Antioxidant capability

●

SLIGHTLY REDUCED
ANTIOXIDANT CAPABILITY

RESULT		ABOUT		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
GPX1	rs1050450	GA	MEDIUM	Predisposition to slightly reduced hydrogen peroxide detoxification and increased oxidative damage.
NQO1	rs1800566	GG	LOW	Predisposition to normal NQO1 activity.
COMT	rs4680	GG	LOW	Predisposition to normal COMT enzyme activity.
SOD2	rs4880	GG	LOW	Predisposition to normal hydrogen peroxide detoxification.
CYP1B1	rs1056836	GG	HIGH	Predisposition to high CYP1B1 activity which could result in accumulation of carcinogenic products.
CYP1A1	rs1048943	TT	LOW	Predisposition to normal CYP1A1 enzyme activity.
GSTP1	rs1695	AG	MEDIUM	Predisposition to slightly reduced GSTP1 activity leading to lower xenobiotic detoxification and increased susceptibility to oxidative stress.

The balance between production and clearance of reactive oxygen species (ROS) is essential for cell survival. Antioxidant cellular systems evolved to maintain a redox homeostasis under different physiological and pathological conditions. Therefore, understanding the status of the antioxidant mechanisms is a key factor for health improvement. The main genes involved in the human antioxidant capability are analysed in this category, allowing us to understand whether we need extra help via specific supplementation or if our internal antioxidant mechanisms work properly.

INDICATIONS

●

NORMAL ANTIOXIDANT CAPABILITY

Normal capacity of metabolizing free radicals and cellular toxins.

●

SLIGHTLY REDUCED ANTIOXIDANT CAPABILITY

Slightly reduced capability of metabolizing free radicals and cellular toxins.

●

REDUCED ANTIOXIDANT CAPABILITY

Reduced capability of metabolizing free radicals and cellular toxins. Increased risk of cellular damage. Prescribe supplementation as suggested at gene level.

●

LOW ANTIOXIDANT CAPABILITY

Low capability of metabolizing free radicals and cellular toxins. High risk of cellular damage. Prescribe supplementation as suggested at gene level.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 10 Intolerance



7. 10. 1. Lactose intolerance risk

RESULT		ABOUT		
LOWER RISK OF LACTOSE INTOLERANCE		Lactose intolerance means that there are insufficient lactase enzymes to break down all the consumed lactose in the intestine. Partially digested or undigested lactose passes into the large intestine and that causes symptoms such as pain, abdominal bloating and diarrhea.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
MCM6	rs182549	CT	LOW	Normal predisposition to lactose tolerance.
MCM6	rs4988235	GA	LOW	Normal predisposition to lactose tolerance.

INDICATIONS

LOWER RISK OF LACTOSE INTOLERANCE

Lower risk of lactose intolerance.

SLIGHTLY INCREASED RISK LACTOSE INTOLERANCE

Slightly increased risk of lactose intolerance. Lower capability to digest lactose. Rather reduce the lactose intake.

MEDIUM-HIGH RISK LACTOSE INTOLERANCE

Medium-high risk of lactose intolerance. Lower capability to digest lactose. Rather reduce or avoid lactose-rich food.

LACTOSE INTOLERANCE

Lactose intolerance. Move to a lactose-free diet.



If your patient suffers from these symptoms and/or has a medium or high risk of developing intolerance is advisable to eliminate as much dairy products from their diet as possible.

Major intestinal symptoms following ingestion of dairy products.

- Nausea
- Abdominal pain
- Spasms
- Swelling and abdominal bloating
- Abdominal gases and flatulence
- Acidic diarrhea
- Vomiting

Other nonspecific symptoms due to an alteration of the intestinal mucosa.

- Abatement
- Tiredness
- Extremities pain
- Skin problems
- Reduced mental concentration
- Nervousness
- Sleep Disorders

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 10 Intolerance



7. 10. 2. Risk of celiac disease

RESULT		ABOUT		
MEDIUM-HIGH RISK OF CELIAC DISEASE		Celiac disease is an autoimmune disorder that occurs in genetically predisposed people where the ingestion of gluten leads to damage in the small intestine and causes digestive problems such as malabsorption of nutrients, abdominal pain or diarrhea. There are different risk haplotypes for celiac disease, the most prevalent is the haplotype HLA-DQ2.5 that covers 90% of celiac disease patients. However, there are other haplotypes (HLA-DQ2.2, HLA-DQ8) which account for 10% of cases and increase the risk of suffering celiac disease. This test determines whether or not an at-risk individual carries this additional risk.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
HLA-DQ2.5	rs2187668	CC	LOW	Normal risk of celiac disease which should be analysed in combination with the other SNPs tested in HLA-DQ region. See the haplotypes in the table below.
HLA-DQ7.5	rs4639334	GG	LOW	Normal risk of celiac disease which should be analysed in combination with the other SNPs tested in HLA-DQ region. See the haplotypes in the table below.
HLA-DQ2.2	rs2395182	GT	MEDIUM	Slightly increased risk of celiac disease which should be analysed in combination with the other SNPs tested in HLA-DQ region. See the haplotypes in the table below.
HLA-DQ2.2	rs4713586	AG	MEDIUM	Slightly increased risk of celiac disease which should be analysed in combination with the other SNPs tested in HLA-DQ region. See the haplotypes in the table below.
HLA-DQ8	rs7454108	TT	LOW	Normal risk of celiac disease which should be analysed in combination with the other SNPs tested in HLA-DQ region. See the haplotypes in the table below.
HLA-DQ2.2	rs7775228	CC	MEDIUM	Slightly increased risk of celiac disease which should be analysed in combination with the other SNPs tested in HLA-DQ region. See the haplotypes in the table below.

INDICATIONS

NO ADDITIONAL RISK OF CELIAC DISEASE No additional risk of celiac disease	LOW RISK OF CELIAC DISEASE Carrier of celiac disease risk variant. Try to reduce the gluten intake (consult your specialist before making any dietary changes).	MEDIUM-HIGH RISK OF CELIAC DISEASE Carrier of celiac disease risk variants. Try to reduce or avoid gluten-containing food (consult your specialist before making any dietary changes).	HIGHER RISK OF CELIAC DISEASE The genetic test indicates a high risk of developing celiac disease. Before initiating any dietary changes, consult your specialist for further analysis.
--	--	---	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 10 Intolerance



7. 10. 2. Risk of celiac disease

HAPLOTYPE	HAPLOTYPE RESULT	HAPLOTYPE SNP DESCRIPTION	HAPLOTYPE RISK
DQ2.5/DQ2.5	Absent	DQ2.5/DQ2.5 = rs2187668 (TT)	HIGH
DQ2.5/DQ2.2	Absent	DQ2.5 = rs2187668 (T) & DQ2.2=rs2395182 (T) + rs7775228 (C) + rs4713586 (A)	HIGH
DQ2.2/DQ2.2	Absent	DQ2.2/DQ2.2=rs2395182 (TT) + rs7775228 (CC) + rs4713586 (AA)	MEDIUM
DQ2.5	Absent	DQ2.5 = rs2187668 (T)	MEDIUM
DQ2.5/DQ8	Absent	DQ2.5= rs2187668 (T) & DQ8= rs7454108 (C)	MEDIUM
DQ2.5/DQ7.5	Absent	DQ2.5= rs2187668 (T) & DQ7.5=rs4639334 (A)	MEDIUM
DQ2.2	Present	DQ2.2=rs2395182 (T) + rs7775228 (C) + rs4713586 (A)	MEDIUM
DQ2.2/DQ8	Absent	DQ2.2 =rs2395182 (T) + rs7775228 (C) + rs4713586 (A) & DQ8= rs7454108 (C)	MEDIUM
DQ2.2/DQ7.5	Absent	DQ2.2 = rs2395182 (T) + rs7775228 (C) + rs4713586 (A) & DQ7.5=rs4639334 (A)	MEDIUM
DQ8/DQ8	Absent	DQ8/DQ8= rs7454108 (CC)	MEDIUM
DQ8/DQ7.5	Absent	DQ8= rs7454108 (C) & DQ7.5=rs4639334 (A)	MEDIUM
DQ8	Absent	DQ8= rs7454108 (C)	MEDIUM
DQ7.5/DQ7.5	Absent	DQ7.5/DQ7.5=rs4639334 (AA)	LOW
DQ7.5	Absent	DQ7.5=rs4639334 (A)	LOW



If your patient suffers from these symptoms and/or has a medium or high risk of developing intolerance is advisable to eliminate as much gluten products from their diet as possible

Major intestinal symptoms
following ingestion of gluten products.

- Bloating
- Abdominal pain
- Skin problems
- Diarrhea, constipation and smelly feces
- Headaches
- Feeling tired
- Unexplained weight loss

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 10 Intolerance



7. 10. 3. Fructose intolerance risk

●

LOWER RISK OF FRUCTOSE INTOLERANCE

Fructose malabsorption, or dietary fructose intolerance, occurs when cells on the surface of the intestines aren't able to break down fructose efficiently. Fructose is a simple sugar, known as a monosaccharide, that comes mostly from fruit and some vegetables. It's also found in honey, agave nectar, and many processed foods that contain added sugars. Symptoms of fructose malabsorption/intolerance include nausea, abdominal pain, diarrhea, vomiting, chronic fatigue, among others.

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ALDOB	rs1800546	CC	LOW	No predisposition to develop hereditary fructose intolerance.
ALDOB	rs76917243	GG	LOW	No predisposition to develop hereditary fructose intolerance.

INDICATIONS

●

LOWER RISK OF FRUCTOSE INTOLERANCE

Lower risk of fructose intolerance.

●

SLIGHTLY INCREASED RISK FRUCTOSE INTOLERANCE

Slightly increased risk of fructose intolerance. Lower capability to digest fructose. Rather reduce the fructose intake.

●

MEDIUM-HIGH RISK FRUCTOSE INTOLERANCE

Medium-high risk of fructose intolerance. Lower capability to digest fructose. Rather reduce or avoid fructose-rich food.

●

HIGH RISK FRUCTOSE INTOLERANCE

Fructose intolerance. Move to a fructose-free diet.



If your patient suffers from these symptoms and/or has a medium or high risk of developing intolerance is advisable to eliminate as much fructose products from their diet as possible

Major intestinal symptoms following ingestion of fructose products.

- Nausea
- Bloating
- Abdominal pain
- Vomiting
- Diarrhea, constipation and smelly feces
- Chronic fatigue
- Malabsorption of certain nutrients, such as iron

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 10 Intolerance



7. 10. 4. Caffeine metabolism

RESULT		ABOUT		
SLOW CAFFEINE METABOLIZER		Metabolism of caffeine. Slower metabolism implies that caffeine will take longer to be degraded and therefore its effects will be more noticeable. However there is a risk of feeling anxious due to excessive consumption. On the other hand, faster metabolism implies that the patient will tend to increase consumption to get the same stimulating effects, since caffeine will be rapidly degraded.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
CYP1A1	rs2470893	CC	HIGH	Predisposition to slow caffeine metabolism.
CYP1A2	rs762551	CA	HIGH	Predisposition to slow caffeine metabolism.

INDICATIONS

FAST CAFFEINE METABOLIZER

Fast speed of caffeine metabolism and increased desire to drink coffee in order to feel the benefits.

INTERMEDIATE-FAST CAFFEINE METABOLIZER

Intermediate speed of caffeine metabolism.

SLOW-INTERMEDIATE CAFFEINE METABOLIZER

Slow caffeine metabolism speed: caffeine will last longer in the body. Be careful with excess caffeine.

SLOW CAFFEINE METABOLIZER

Very slow caffeine metabolism speed: caffeine will last longer in the body. Be careful with excess caffeine.



If your patient suffers from these symptoms and/or has a medium or high risk of developing intolerance is advisable to eliminate as much caffeine products from their diet as possible.

Major intestinal symptoms following ingestion of caffeine products.

- ▶ Headaches
- ▶ Acing heartbeat
- ▶ Jitters
- ▶ Nervousness or anxiousness
- ▶ Restlessness
- ▶ Insomnia

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 10 Intolerance



7. 10. 5. Alcohol metabolism

RESULT		ABOUT		
NORMAL ALCOHOL METABOLISM		People of certain genetic type may experience symptoms like redness or flushing of the face and neck after consuming alcohol. These reactions can result from variants in the ALDH2 gene which is involved in breaking down alcohol.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ALDH2	rs671	GG	LOW	Predisposition to normal alcohol metabolism.

INDICATIONS

NORMAL ALCOHOL METABOLISM

Normal risk of alcohol toxicity due to a normal metabolism.

NORMAL-INTERMEDIATE ALCOHOL METABOLISM

Moderate risk of alcohol toxicity due to a slightly slower metabolism.

INTERMEDIATE-SLOW ALCOHOL METABOLISM

Medium-high risk of alcohol toxicity due to slow metabolism.

SLOW ALCOHOL METABOLISM

High risk of alcohol toxicity due to slow metabolism.



If your patient suffers from these symptoms and/or has a medium or high risk of developing intolerance is advisable to eliminate as much alcohol products from their diet as possible.

Major intestinal symptoms following ingestion of alcohol products.

- ▶ Vomiting
- ▶ Facial redness (flushing)
- ▶ Red, itchy skin bumps (hives)
- ▶ Worsening of pre-existing asthma
- ▶ Runny or stuffy nose
- ▶ Low blood pressure
- ▶ Nausea and vomiting

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 11 Hormones



7. 11. 1. Leptin

●

HIGH RISK OF DECREASED LEPTIN LEVELS

RESULT		ABOUT		
		Leptin is a hormone which main function is sending a signal to the brain for food intake regulation. Leptin is commonly called the "satiety hormone". Low levels of leptin may imply problems of overeating and/or burning the stored fat. LEP-R is the gene coding for the cellular receptor of the leptin hormone. Its capability to bind leptin and start the cellular signalling is key for the satiety regulation function. Lower leptin binding capability may lead to high possibilities of leptin resistance, overeating and lower fat burning.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
LEP	rs7799039	GA	HIGH	Predisposition to increased levels of leptin leading to leptin receptor saturation that may lead to increased risk of overeating and lower fat burning.

INDICATIONS

●

LOW RISK OF DECREASED LEPTIN LEVELS

Low Risk Of Decreased Leptin Levels

●

MEDIUM-LOW RISK OF DECREASED LEPTIN LEVELS

Medium-Low Risk Of Decreased Leptin Levels

●

MEDIUM-HIGH RISK OF DECREASED LEPTIN LEVELS

Medium-High Risk Of Decreased Leptin Levels

●

HIGH RISK OF DECREASED LEPTIN LEVELS

High Risk Of Decreased Leptin Levels

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 11 Hormones



7. 11. 2. Visfatin

RESULT		ABOUT		
HIGH RISK OF INCREASED VISFATIN LEVELS		Visfatin is an adipokine with an inflammatory and catabolic profile that has been associated with several metabolic risk factors, such as obesity, insulin resistance, and Type-II diabetes.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
NAMPT	rs9770242	AA	HIGH	High predisposition to increased levels of circulating visfatin augmenting considerably the risk of an altered inflammatory response.

INDICATIONS

LOW RISK OF INCREASED VISFATIN LEVELS

Low Risk Of Increased Visfatin Levels

MEDIUM-LOW RISK OF INCREASED VISFATIN LEVELS

Medium-Low Risk Of Increased Visfatin Levels

MEDIUM-HIGH RISK OF INCREASED VISFATIN LEVELS

Medium-High Risk Of Increased Visfatin Levels

HIGH RISK OF INCREASED VISFATIN LEVELS

High Risk Of Increased Visfatin Levels

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 11 Hormones



7. 11. 3. Ghrelin

RESULT		ABOUT		
HIGH GHRELIN RECEPTOR (GHSR) EXPRESSION		Ghrelin is a hormone produced in the gut, often termed "the hunger hormone", since it causes an increase in appetite through its effect in the brain. Imbalances in ghrelin are associated with appetite increase, increased calorie consumption and fat storage.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
GHSR	rs490683	GG	HIGH	Predisposition to normal ghrelin receptor (GHSR) expression. Circulating levels of this hormone and the sensation of hunger are increased.

INDICATIONS

LOW GHRELIN RECEPTOR (GHSR) EXPRESSION Low Ghrelin Receptor (Ghsr) Expression	LOW-INTERMEDIATE GHRELIN RECEPTOR (GHSR) EXPRESSION Low-intermediate ghrelin receptor (ghsr) expression	INTERMEDIATE- HIGH GHRELIN RECEPTOR (GHSR) EXPRESSION Intermediate- High Ghrelin Receptor (Ghsr) Expression	HIGH GHRELIN RECEPTOR (GHSR) EXPRESSION High Ghrelin Receptor (GHSR) Expression
--	--	--	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 11 Hormones



7. 11. 4. Adiponectin

RESULT		ABOUT		
HIGH RISK OF DECREASED ADIPONECTIN LEVELS		Adiponectin is a hormone that regulates glucose levels and fatty acid breakdown. Low levels of adiponectin are associated with inflammation, lipid abnormalities and insulin resistance.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ADIPOQ	rs1501299	GG	HIGH	High predisposition to lower adiponectin plasma levels that leads to an increased inflammation process, lipid abnormalities and insulin resistance.
ADIPOQ	rs2241766	TT	HIGH	High predisposition to lower adiponectin plasma levels that leads to an increased inflammation process, lipid abnormalities and insulin resistance.
INDICATIONS				
LOW RISK OF DECREASED ADIPONECTIN LEVELS Low Risk Of Decreased Adiponectin Levels	MEDIUM-LOW RISK OF DECREASED ADIPONECTIN LEVELS Medium-Low Risk Of Decreased Adiponectin Levels	MEDIUM-HIGH RISK OF DECREASED ADIPONECTIN LEVELS Medium-High Risk Of Decreased Adiponectin Levels	HIGH RISK OF DECREASED ADIPONECTIN LEVELS High Risk Of Decreased Adiponectin Levels	

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 12 Inflammation



7. 12. 1. TNF-α

RESULT		ABOUT		
LOW RISK OF DYSREGULATED TNF-α LEVELS.		TNF-α is a pro-inflammatory cytokine, strongly linked to many inflammatory conditions, expressed in, and secreted by adipose tissues. Increased levels are associated with obesity-induced inflammation, adiposity and insulin resistance.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
TNF-α	rs1800629	GG	LOW	Predisposition to average levels of TNF-alpha.

INDICATIONS

LOW RISK OF DYSREGULATED TNF-α LEVELS.

Low Risk Of Dysregulated Tnf-A Levels.

MEDIUM-LOW RISK OF INCREASED TNF-α LEVELS

Medium-Low Risk Of Increased TNF-α Levels

MEDIUM-HIGH RISK OF INCREASED TNF-α LEVELS

Medium-High Risk Of Increased TNF-α Levels

HIGH RISK OF INCREASED TNF-α LEVELS

High Risk Of Increased TNF-α Levels

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 12 Inflammation



7. 12. 2. IL-6

●

HIGH RISK OF INCREASED IL-6 LEVELS

IL-6 is an interleukin with mainly pro-inflammatory functions and is commonly used as inflammatory marker. High levels of IL-6 are associated with obesity, insulin resistance and metabolic syndrome.

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
IL-6	rs1800795	GG	HIGH	Predisposition to highly increased levels of IL-6. Pro-inflammation.

INDICATIONS

●

LOW RISK OF INCREASED IL-6 LEVELS

Low Risk Of Increased Il-6 Levels

●

MEDIUM-LOW RISK OF INCREASED IL-6 LEVELS

Medium-Low Risk Of Increased Il-6 Levels

●

MEDIUM-HIGH RISK OF INCREASED IL-6 LEVELS

Medium-High Risk Of Increased Il-6 Levels

●

HIGH RISK OF INCREASED IL-6 LEVELS

High Risk Of Increased Il-6 Levels

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 12 Inflammation



7. 12. 3. IL-10

RESULT		ABOUT		
LOW RISK OF DECREASED ANTIINFLAMMATORY CYTOKINE IL-10 LEVELS		IL-10 is a cytokine with potent anti-inflammatory properties.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
IL-10	rs1800896	CC	LOW	Predisposition to higher levels of the anti-inflammatory cytokine IL-10.

INDICATIONS

LOW RISK OF DECREASED ANTIINFLAMMATORY CYTOKINE IL-10 LEVELS Low Risk Of Decreased Antiinflammatory Cytokine Il-10 Levels	MEDIUM-LOW RISK OF DECREASED ANTIINFLAMMATORY CYTOKINE IL-10 LEVELS Medium-Low Risk Of Decreased Antiinflammatory Cytokine Il-10 Levels	MEDIUM-HIGH RISK OF DECREASED ANTIINFLAMMATORY CYTOKINE IL-10 LEVELS Medium-High Risk Of Decreased Antiinflammatory Cytokine Il-10 Levels	HIGH RISK OF DECREASED ANTIINFLAMMATORY CYTOKINE IL-10 LEVELS High Risk Of Decreased Antiinflammatory Cytokine Il-10 Levels
--	--	--	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

A

7. 13. 1. Vitamin A

RESULT		ABOUT		
LOW RISK OF VITAMIN A DEFICIENCY		Inherited risk of vitamin A metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
BCMO1	rs12934922	AT	MEDIUM	Increased predisposition to reduced provitamin A conversion and increased fasting β -carotene concentrations.
BCMO1	rs7501331	CC	LOW	Normal risk of vitamin A deficiency.
INDICATIONS				
LOW RISK OF VITAMIN A DEFICIENCY Normal vitamin A metabolism. Ensure daily recommended intake.	MEDIUM-LOW RISK OF VITAMIN A DEFICIENCY Low risk of vitamin A deficiency. Ensure daily recommended intake or slightly increase it.	MEDIUM-HIGH RISK OF VITAMIN A DEFICIENCY Medium risk of vitamin A deficiency. Ensure daily recommended intake and increase it. Supplementation should be evaluated.	HIGH RISK OF VITAMIN A DEFICIENCY High risk of vitamin A deficiency. Increase daily vitamin A intake. Supplementation should be evaluated.	

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

B6

7. 13. 2. Vitamin B6

RESULT		ABOUT		
HIGH RISK OF VITAMIN B6 DEFICIENCY		Inherited risk of vitamin B6 metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
NBPF3	rs4654748	CC	HIGH	High risk of low plasma vitamin B6 concentrations.

INDICATIONS

LOW RISK OF VITAMIN B6 DEFICIENCY

Normal vitamin B6 metabolism.
Ensure daily recommended intake.

MODERATE RISK OF VITAMIN B6 DEFICIENCY

Little predisposition to a vitamin B6 deficiency. Make sure that the recommended daily intake is met.

MEDIUM-HIGH RISK OF VITAMIN B6 DEFICIENCY

Medium risk of vitamin B6 deficiency. Ensure daily recommended intake and increase it. Supplementation should be evaluated.

HIGH RISK OF VITAMIN B6 DEFICIENCY

High risk of vitamin B6 deficiency. Increase daily vitamin B6 intake. Supplementation should be evaluated.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

B9

7. 13. 3. Vitamin B9 (folate)

RESULT		ABOUT		
MEDIUM-LOW RISK OF VITAMIN B9 (Folate) DEFICIENCY		Inherited risk of vitamin B9 (folate) metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
MTHFR	rs1801133	GA	MEDIUM	Increased risk of lower serum levels of folate.

INDICATIONS

LOW RISK OF VITAMIN B9 (Folate) DEFICIENCY

Normal folate metabolism. Ensure daily recommended intake.

MEDIUM-LOW RISK OF VITAMIN B9 (Folate) DEFICIENCY

Low risk of folate deficiency. Ensure daily recommended intake.

MEDIUM-HIGH RISK OF VITAMIN B9 (Folate) DEFICIENCY

Medium risk of folate deficiency. Ensure daily recommended intake. It is recommended to supplement with L-methylfolate due to a lower capability to activate folate. It also impacts lower B12 levels when low levels of folate are active.

HIGH RISK OF VITAMIN B9 (Folate) DEFICIENCY

High risk of folate deficiency. Ensure daily recommended intake. Highly recommended to supplement with L-methylfolate due to a almost null capability to activate folate. It also impacts lower B12 levels when low levels of folate are active.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

B12

7. 13. 4. Vitamin B12

RESULT		ABOUT		
MEDIUM-LOW RISK OF VITAMIN B12 DEFICIENCY		Inherited risk of vitamin B12 metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
FUT2	rs602662	GA	MEDIUM	Increased risk of vitamin B12 deficiency.

INDICATIONS

LOW RISK OF VITAMIN B12 DEFICIENCY

Normal vitamin B12 metabolism.
Ensure daily recommended intake.

MEDIUM-LOW RISK OF VITAMIN B12 DEFICIENCY

Low risk of vitamin B12 deficiency.
Ensure daily recommended intake.

MEDIUM-HIGH RISK OF VITAMIN B12 DEFICIENCY

Medium risk of vitamin B12 deficiency.
Ensure daily recommended intake and increase it. Supplementation should be evaluated.

HIGH RISK OF VITAMIN B12 DEFICIENCY

High risk of vitamin B12 deficiency.
Increase daily vitamin B12 intake. Supplementation should be evaluated.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

C

7. 13. 5. Vitamin C

RESULT		ABOUT		
LOW RISK OF VITAMIN C DEFICIENCY		Inherited risk of vitamin C metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
SLC23A2	rs1279683	GA	LOW	Normal risk of vitamin C deficiency.
SLC23A1	rs33972313	CC	LOW	Normal risk of vitamin C deficiency.
INDICATIONS				
LOW RISK OF VITAMIN C DEFICIENCY Normal vitamin C metabolism and levels. Ensure daily recommended intake.	MEDIUM-LOW RISK OF VITAMIN C DEFICIENCY Low risk of Viamin C deficiency. Ensure daily recommended intake.	MEDIUM-HIGH RISK OF VITAMIN C DEFICIENCY Medium risk of Vitamin C deficiency. Ensure daily recommended intake. Supplementation strategies might be of interest.	HIGH RISK OF VITAMIN C DEFICIENCY High risk of Vitamin C deficiency. Ensure daily recommended intake. Supplementation strategies would be recommended.	

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

D

7. 13. 6. Vitamin D

RESULT		ABOUT		
MEDIUM-LOW RISK OF VITAMIN D DEFICIENCY		Inherited risk of vitamin D metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
GC	rs2282679	TT	LOW	Normal risk of vitamin D deficiency.
CYP2R1	rs10741657	GG	HIGH	High risk of low serum levels of vitamin D.
NADSYN1	rs12785878	TT	HIGH	High risk of low serum levels of vitamin D.
CYP2R1	rs2060793	GG	LOW	Normal risk of vitamin D deficiency.
NADSYN1	rs3829251	GG	LOW	Normal risk of vitamin D deficiency.

INDICATIONS

LOW RISK OF VITAMIN D DEFICIENCY

Normal vitamin D metabolism and levels. Ensure daily recommended intake.

MEDIUM-LOW RISK OF VITAMIN D DEFICIENCY

Low risk of Viamin D deficiency. Ensure daily recommended intake.

MEDIUM-HIGH RISK OF VITAMIN D DEFICIENCY

Medium risk of Vitamin D deficiency. Ensure daily recommended intake. Supplementation strategies might be of interest.

HIGH RISK OF VITAMIN D DEFICIENCY

High risk of Vitamin D deficiency. Ensure daily recommended intake. Supplementation strategies would be recommended.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 13 Vitamins

E

7. 13. 7. Vitamin E

RESULT		ABOUT		
HIGH RISK OF VITAMIN E DEFICIENCY		Inherited risk of vitamin E metabolism deficiency or low plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
INTERGENIC	rs12272004	CC	HIGH	High risk of low plasma levels of alpha-tocopherol (Vitamin E).
ZPR1	rs964184	CC	HIGH	High risk of low plasma levels of alpha-tocopherol (Vitamin E).

INDICATIONS

LOW RISK OF VITAMIN E DEFICIENCY

Normal vitamin E metabolism and levels. Ensure daily recommended intake.

MEDIUM-LOW RISK OF VITAMIN E DEFICIENCY

Low risk of Viamin E deficiency. Ensure daily recommended intake.

MEDIUM-HIGH RISK OF VITAMIN E DEFICIENCY

Medium risk of Vitamin E deficiency. Ensure daily recommended intake. Supplementation strategies might be of interest.

HIGH RISK OF VITAMIN E DEFICIENCY

High risk of Vitamin E deficiency. Ensure daily recommended intake. Supplementation strategies would be recommended.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 1. Calcium malabsorption risk

RESULT		ABOUT		
LOW RISK OF CALCIUM MALABSORPTION		Calcium dissolves in the stomach and is absorbed through the lining of the small intestine into the blood stream. Once in the blood stream, calcium builds bone, regulates the expansion and contraction of the blood vessels, and performs other important functions. Common factors for calcium malabsorption are a diet high in phytic acid (present in wholegrains), high levels of sodium intake, smoking and also genetic factors related to Vitamin D. In this category, the genetic factors related to a predisposition to calcium malabsorption due to lower levels of 25(OH) D (Vitamin D) are analysed. Therefore, a high risk of malabsorption would require an increase in vitamin D consumption or even controlled supplementation.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
CYP2R1	rs10766197	GA	MEDIUM	Predisposition to slightly reduced vitamin D levels and calcium absorption.
GC	rs2282679	TT	LOW	Predisposition to normal vitamin D levels and calcium absorption.

INDICATIONS

LOW RISK OF CALCIUM MALABSORPTION
Low inherited risk of calcium malabsorption.

MEDIUM-LOW RISK OF CALCIUM MALABSORPTION
Medium-low inherited risk of calcium malabsorption.

MEDIUM-HIGH RISK OF CALCIUM MALABSORPTION
Medium-high inherited risk of calcium malabsorption.

HIGH RISK OF CALCIUM MALABSORPTION
High inherited risk of calcium malabsorption.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 2. Predisposition to dysregulated calcium levels

RESULT		ABOUT		
SLIGHTLY INCREASED RISK OF DYSREGULATED PLASMA CALCIUM LEVELS		The predisposition to low or high levels of plasma calcium are analyzed in this category. A predisposition to high levels of calcium and increased absorption would be a warning against calcium supplementation due to the potential increased risk of vascular calcification.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
DGKD	rs1550532	GG	LOW	Predisposition to normal serum levels of calcium.
CYP24A1	rs1570669	AA	HIGH	Predisposition to reduced serum calcium levels and bone mineral density.
CASR	rs17251221	AA	LOW	Predisposition to normal serum calcium levels.
CASR	rs1801725	GG	LOW	Predisposition to normal serum calcium levels.
CARS	rs7481584	GA	MEDIUM	Predisposition to slightly reduced serum calcium levels.
GCKR	rs780094	TC	MEDIUM	Predisposition to slightly reduced serum calcium levels.

INDICATIONS

NO ADDITIONAL RISK OF DYSREGULATED PLASMA CALCIUM LEVELS

No additional risk of dysregulated plasma calcium levels.

SLIGHTLY INCREASED RISK OF DYSREGULATED PLASMA CALCIUM LEVELS

Slightly increased risk of dysregulated plasma calcium levels.

INCREASED RISK OF DYSREGULATED PLASMA CALCIUM LEVELS

Increased risk of dysregulated plasma calcium levels.

HIGHER RISK OF DYSREGULATED PLASMA CALCIUM LEVELS

High risk of dysregulated plasma calcium levels.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 3. Risk of iron overload

RESULT		ABOUT		
LOW RISK OF HEMOCHROMATOSIS		Iron overload is defined as excess stores of iron in the body. Excess iron is deposited in organs throughout the body. The most notable organs with iron deposition are the liver, heart, and endocrine glands. Resulting symptoms and diseases are related to specific organ damage. In this category, the genetic risk of iron overload on high intake is analysed.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
HFE	rs1800562	GG	LOW	Predisposition to normal absorption of dietary iron.

INDICATIONS

LOW RISK OF HEMOCHROMATOSIS No additional risk of iron overload.	MEDIUM-LOW RISK OF HEMATOCHROMATOSIS Some risk of having increased iron absorption on high iron intake. Avoid iron excess.	MEDIUM-HIGH RISK O FHEMATOCHROMATOSIS Medium risk of having increased iron absorption on high iron intake. Avoid iron excess and/or supplements.	HIGH RISK OF HEMATOCHROMATOSIS High risk of having increased iron absorption on high iron intake. Avoid iron excess and/or supplements.
---	---	---	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 4. Risk of low iron plasma levels

RESULT		ABOUT		
LOW RISK OF DECREASED IRON LEVELS		Low iron levels may lead to anemia. In this category, the genetic risk of low transference of iron into the body is analysed. When your body has a predisposition to low iron levels, it will be necessary to ensure a diet with proper levels of iron.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
TF	rs3811647	GG	LOW	Predisposition to normal serum ferritin and iron levels.
TMPRSS6	rs4820268	GA	MEDIUM	Predisposition to slightly reduced iron levels.
TF	rs8177253	CC	LOW	Predisposition to normal iron binding capacity.

INDICATIONS

LOW RISK OF DECREASED IRON LEVELS No additional inherited risk of low iron levels.	MEDIUM-LOW RISK OF DECREASED IRON LEVELS Some risk of having lower iron transference, only when iron intake is low. Ensure dietary daily recommended intake.	MEDIUM-HIGH RISK OF DECREASED IRON LEVELS Moderate risk of having lower iron transference, only when iron intake is low. In that case, supplementation would be recommended.	HIGH RISK OF DECREASED IRON LEVELS High risk of having lower iron transference, only when iron intake is low. In that case, supplementation would be recommended.
---	---	---	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 5. Predisposition to dysregulated magnesium levels

RESULT		ABOUT		
MEDIUM-HIGH RISK OF DYSREGULATED MAGNESIUM LEVELS		Inherited risk of low magnesium plasma levels.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
CASR	rs17251221	AA	LOW	Predisposition to normal serum magnesium levels.
TRPM6	rs11144134	TT	HIGH	Predisposition to lower serum magnesium levels.
SHROOM3	rs13146355	GG	HIGH	Predisposition to lower serum magnesium levels.
DCDC5	rs3925584	TT	LOW	Predisposition to normal serum magnesium levels.
MUC1	rs4072037	CT	HIGH	Increased risk of decreased serum magnesium levels associated with lower kidney function measure (eGFR).

INDICATIONS

NO ADDITIONAL RISK OF DYSREGULATED MAGNESIUM LEVELS

No additional risk of dysregulated plasma magnesium levels.

MEDIUM-LOW RISK OF DYSREGULATED MAGNESIUM LEVELS

Some risk of dysregulated plasma magnesium levels.

MEDIUM-HIGH RISK OF DYSREGULATED MAGNESIUM LEVELS

Medium risk of dysregulated plasma magnesium levels.

HIGH RISK OF DYSREGULATED MAGNESIUM LEVELS

High risk of dysregulated plasma magnesium levels.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 6. Predisposition to dysregulated selenium levels

RESULT		ABOUT		
NO ADDITIONAL RISK OF DYSREGULATED SELENIUM LEVELS		Selenium is an essential mineral and micronutrient. It is fundamental to human health and found in many foods. It is found in meat, grain cereals, egg yolk, milk, brazil nuts, mushrooms, garlic and seafood (hence, selenium levels are high in populations with high intake of seafood). Understanding the predisposition to low or high selenium levels will help for ensuring the proper selenium daily intake.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
AGA	rs1395479	CC	LOW	Predisposition to normal selenium levels.
SLC39A11	rs891684	GG	LOW	Predisposition to normal serum selenium levels.

INDICATIONS

NO ADDITIONAL RISK OF DYSREGULATED SELENIUM LEVELS No additional risk of dysregulated plasma selenium levels.	MEDIUM-LOW RISK OF DYSREGULATED SELENIUM LEVELS Some risk of dysregulated plasma selenium levels.	MEDIUM-HIGH RISK OF DYSREGULATED SELENIUM LEVELS Medium risk of dysregulated plasma selenium levels.	HIGH RISK OF DYSREGULATED SELENIUM LEVELS High risk of dysregulated plasma selenium levels.
--	--	---	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 14 Minerals



7. 14. 7. Sodium sensitivity

RESULT		ABOUT		
MEDIUM-LOW SODIUM SENSITIVITY		Inherited risk of dietary salt-induced blood pressure.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
ACE	rs4343	GA	MEDIUM	Predisposition to increased sodium sensitivity associated with increased risk of sodium sensitive hypertension.

INDICATIONS

LOW SODIUM SENSITIVITY

Normal sodium sensitivity: no increased blood pressure risk due to salt consumption.

MEDIUM-LOW SODIUM SENSITIVITY

Slightly increased sodium sensitivity: moderately increased blood pressure risk due to salt consumption.

MEDIUM-HIGH SODIUM SENSITIVITY

Moderate sodium sensitivity: increased blood pressure risk due to salt consumption. Reduce current salt consumption, if daily intake is high.

HIGH SODIUM SENSITIVITY

High sodium sensitivity: high blood pressure risk due to salt consumption. Reduce current salt consumption, if daily intake is high.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 15 Effectiveness of diets



7. 15. 1. Efficacy of low calorie diets

RESULT

ABOUT

●

VERY LOW EXPECTED BENEFIT FROM LOW-CALORIE DIET

A complete set of genes related to the expected efficacy of a low-calorie diet is analysed in this category.

GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARG	rs1801282	CC	HIGH	No predisposition to weight loss induced by a low calorie diet.
ADIPOQ	rs17300539	GG	HIGH	No predisposition to weight loss induced by a low calorie diet.
LEPR	rs1805134	TT	HIGH	No predisposition to weight loss induced by a low calorie diet.
ACSL5	rs2419621	CC	HIGH	No predisposition to weight loss induced by a low calorie diet.
ADRB2	rs1042714	GC	MEDIUM	Increased predisposition to weight loss induced by a low calorie diet.

INDICATIONS

●

VERY LOW EXPECTED BENEFIT FROM LOW-CALORIE DIET

A pure low-calorie diet may not be the best option for weight loss.

●

MEDIUM-LOW EXPECTED BENEFIT FROM LOW-CALORIE DIET

A pure low-calorie diet may not be the best option for weight loss. However, a reduction in calorie intake may be beneficial.

●

MEDIUM-HIGH EXPECTED BENEFIT FROM LOW-CALORIE DIET

A low-calorie diet may be one of the best options for weight loss. Try to dramatically reduce calorie intake.

●

HIGH EXPECTED BENEFIT FROM LOW-CALORIE DIET

High expected efficacy of a low-calorie diet. It is strongly recommended to follow it.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 15 Effectiveness of diets



7. 15. 2. Efficacy of low carbohydrate diets

RESULT		ABOUT		
MEDIUM-HIGH EXPECTED BENEFIT FROM LOW-CARBOHYDRATE DIET		A complete set of genes related to the expected efficacy of a low-carbohydrate diet is analysed in this category.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
KCTD10	rs10850219	GG	LOW	Predisposition to weight loss induced by a low carbohydrate diet.
MMAB	rs2241201	CG	HIGH	No predisposition to weight loss induced by a low carbohydrate diet.

INDICATIONS

VERY LOW EXPECTED BENEFIT FROM LOW-CARBOHYDRATE DIET A pure low-carbohydrate diet may not be the best option for weight loss.	MEDIUM-LOW EXPECTED BENEFIT FROM LOW-CARBOHYDRATES DIET A pure low-carbohydrate diet may not be the best option for weight loss. However, a reduction in carbohydrate intake may be beneficial.	MEDIUM-HIGH EXPECTED BENEFIT FROM LOW-CARBOHYDRATE DIET A low-carbohydrate diet may be one of the best option for weight loss. Try to dramatically reduce carbohydrate intake.	HIGH EXPECTED BENEFIT FROM LOW-CARBOHYDRATE DIET High expected efficacy of a low-carbohydrate diet. It is strongly recommended to follow it.
---	---	--	--

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Complete genetic results

7

7. 15 Effectiveness of diets



7. 15. 3. Efficacy of low fat diets

RESULT		ABOUT		
MEDIUM-LOW EXPECTED BENEFIT FROM LOW-FAT DIET		A complete set of genes related to the expected efficacy of a low-fat diet is analysed in this category.		
GENE	SNP	GENOTYPE	RISK	DESCRIPTION
PPARG	rs1801282	CC	HIGH	No predisposition to weight loss induced by a low fat diet.
GHSR	rs490683	GG	HIGH	No predisposition to weight loss induced by a low fat diet. Also applicable after gastric bypass.
APOA2	rs5082	GG	LOW	Predisposition to weight loss induced by a low fat diet.
SH2B1	rs7498665	AG	MEDIUM	Increased predisposition to weight loss induced by a low fat diet.
TCF7L2	rs7903146	CT	MEDIUM	Increased predisposition to weight loss induced by a low fat diet.
FTO	rs9939609	TA	MEDIUM	Increased predisposition to weight loss induced by a low fat diet.
INDICATIONS				
VERY LOW EXPECTED BENEFIT FROM LOW-FAT DIET A pure low-fat diet may not be the best option for weight loss.	MEDIUM-LOW EXPECTED BENEFIT FROM LOW-FAT DIET A pure low-fat diet may not be the best option for weight loss. However, a reduction of fat intake may be beneficial.	MEDIUM-HIGH EXPECTED BENEFIT FROM LOW-FAT DIET A low-fat diet may be one of the best options for weight loss. Try to dramatically reducefat intake.	HIGH EXPECTED BENEFIT FROM LOW-FAT DIET The expected efficacy of a low-fat diet is high. It is strongly recommended to follow it.	

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Methodology

8

How were the genetic variants selected and evaluated?

This test was developed by a multidisciplinary team of medical doctors, geneticists, and programmers, following highest quality standards. In particular, an expert team specialized in the curation of genetic variants reviewed each variant to ensure that selection, interpretation and impact of variants in the algorithms are based on the highest scientific evidence.

The following selection criteria were applied for classifying genetic variants:

Level 1A: Annotation for a variant in medical society-endorsed or implemented in a major health system.

Level 1B: Annotation for a variant where the preponderance of evidence shows an association. The association must be replicated in more than one cohort with significant p-values, and preferably will have a strong effect size.

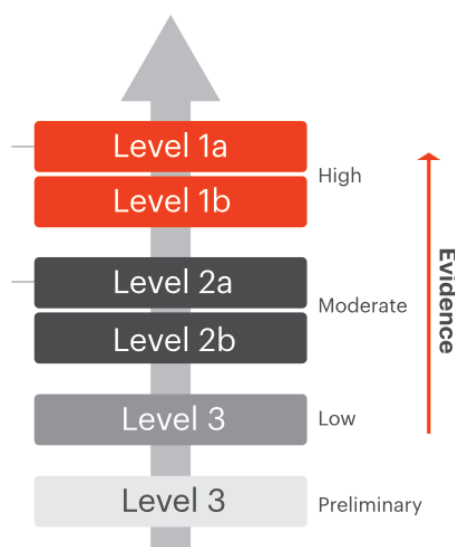
Level 2A: Annotation for a variant that qualifies for level 2B where the variant is within a Very Important known gene, so functional significance is more likely.

Level 2B: Annotation for a variant with moderate evidence of an association. The association must be replicated but there may be some studies that do not show statistical significance, and/or the effect size may be small.

Level 3: Annotation for a variant based on a single significant (not yet replicated) study or annotation for a variant evaluated in multiple studies but lacking clear evidence of an association.

Level 4: Annotation based on a case report, non-significant study or in vitro, molecular or functional assay evidence only.

Only genetic variants from level 1A to 2A were selected.



How was it analyzed?

The DNA was extracted from the buccal swab sample you provided and was analyzed by our clinical analysis laboratory. DNA was extracted using the KingFisher Flex® robotic extraction system (Thermo Fisher Scientific). The study of the genetic variants was performed by NGS (Next Generation Sequencing) using the Ion GeneStudio S5 system (Thermo Fisher Scientific).

References

9



References

Scan the QR code to access our NutriGen report reference page.

* These recommendations are based only in the analysis of your genetic test. Always seek the advice of your physician or other qualified health specialist before proceeding with any nutritional or dietary modifications.

Together
we create the future
of personalizing medicine.

